Social Learning, Behavioral Biases and Group Outcomes *

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Abstract

While evidence shows that cognitive biases influence individual decisions, economic outcomes often result from interactions among individuals, such as through social learning. This paper investigates the impact of social learning on a broad range of behavioral biases, reflecting economically relevant settings. Experimental evidence shows how social learning can reduce or amplify errors stemming from behavioral biases, affecting group outcomes. This suggests that social learning does not systematically eliminate biases, and can either mitigate or exacerbate their impact in settings such as the interpretation of statistical information or investment decisions.

Keywords— *Social Learning, Group Outcomes, Biases, Experiment, Relative Confidence*
JEL Codes: D91, C91, D83

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1 Introduction

Ample evidence documents that behavioral biases are omnipresent in individual decision-making and belief formation. While these biases are extensively studied at the individual level, economics is often concerned with group outcomes, shaped by the interactions of multiple agents. Moreover, many economic contexts are inherently social, allowing individuals to learn from others, which may attenuate or amplify the role of behavioral biases. For instance, we turn to peers for guidance on crucial decisions, such as educational or investment choices, and consult strangers on social media to interpret political developments, economic data, or public health statistics. This raises the question of whether and how observing and learning from others — social learning — can mitigate or amplify biases in the aggregate.

Consider a group of retail investors forming beliefs about the quality of a newly appointed CEO, who has been in office for one month. Over recent weeks, the company's stock price has declined. However, the broader market sector has experienced even sharper losses during the same period. Some investors appropriately account for sector-wide movements and attribute little of the price change to the CEO's performance. Others, by contrast, overreact to the declining stock price, inferring poor leadership without isolating firm-specific factors. After forming their individual assessments, investors begin sharing their views with one another. This process of social learning may either exacerbate initial overreactions or help correct them.

Building on the idea emerging from the example, this paper investigates the extent to which social learning can amplify or curb errors caused by a variety of economically relevant behavioral biases, and how this impacts group outcomes, that is the prevalence of the bias in a group. The answer to this question is prima facie unclear. The reason is that classic social learning settings in economics presume that people know that those they observe have information that they do not have. However, in the context of overcoming cognitive biases, individuals might not realize that even conditional on the same information, others may be better at solving a problem than them. As a result, they might disregard others' actions as mistakes if they differ from their own, reducing the potential of social learning to mitigate biases. In fact, if unbiased individuals are less confident in their decisions, they may imitate the behavior of biased ones, amplifying the prevalence of mistakes in groups.

To illustrate this point, I develop a conceptual framework where agents choose actions, observe others' decisions on identical tasks, and decide whether to revise their initial choices. Agents are prone to mistakes, and this is common knowledge; they will hold beliefs concerning the optimality of their own and of the observed actions. In principle, agents differ in their probability of committing mistakes, that is, they have different levels of performance. The framework incorporates the concept of relative confidence as a regulator of social learning behavior. Relative confidence measures how much more confident an agent is in their own action compared to others'. This drives whether individuals imitate or dismiss observed decisions. In other words, relative confidence can be thought of as the difference between confidence in one's own action and confidence in the observed action. A key assumption is that, in any given task, relative confidence is structurally related to relative performance. Crucially, in this framework, when the correlation between relative confidence and relative performance is negative, social learning will increase the group bias and vice versa. The goal of the framework is to convey the key intuitions formally and to derive clear predictions for the experimental investigation.

To test these ideas, I conduct preregistered online experiments where participants solve cognitive tasks that reflect well-documented biases. Unlike traditional social learning experiments, these tasks do not feature private information, focusing instead on participants' beliefs about others' capacities. Specifically, I study the following ten biases: *failure to condition on contingencies* (AC), *correlation neglect* (CN), *following misleading intuition* (CRT), *exponential growth bias* (EGB), *failure in constrained optimization* (KS), *1/N heuristic* (PC), *gambler's fallacy* (GF), *sample size neglect* (SSN), *failure to account for noise* (RM), *thinking about average instead of marginal costs/benefits* (TM).[1,2] In the *Baseline* condition, each task is characterized by five steps, in which participants: (i) provide their answer to the task, (ii) provide their confidence in their answer, (iii) are exposed to another participant's answer to the exact same task, (iv) provide their assessment of the optimality probability of the other participant's answer, and (v) have the opportunity to change their initial answer. Relative confidence is constructed as the difference between the quantities elicited in steps (ii) and (iv).

The experimental setup I illustrate differs in two key aspects from the canonical social learning experiments in the literature, mainly inspired by Anderson and Holt's (1997) influential paper. First, I employ different tasks, each with a correct solution that can be reached with the provided information. In paradigms à la Anderson and Holt (1997), the task is one in which participants observe a private, noisy signal about some unobservable state and then sequentially choose the state they believe to be true. Second, none of the tasks that I selected feature *external uncertainty*: once the problem is correctly understood, there is a determinate correct answer. This has an important implication. In canonical social learning experiments, participants know that others possess potentially valuable private information. In the environments studied here, by contrast, all participants face the same problem, and social learning can only be valuable because others may understand that common problem better. Participants may fail to recognize

[1]The fact that these cognitive biases play a relevant role in economic and financial decisions is largely documented in the literature. The early finding that investors typically do not sufficiently diversify, is explained by correlation neglect (Chinco et al., 2022; Laudenbach et al., 2022). Also, even when investors diversify, a relevant portion of them employ the 1/N heuristics in building their portfolios (Benartzi and Thaler, 2001). Stango and Zinman (2009) show how exponential growth bias accounts for sub-optimal saving behavior, accounting for other relevant factors including financial sophistication. The influence of sample size neglect and gambler's fallacy has been documented in betting (e.g. Camerer, 1989) and financial markets (Baquero, 2006). Frederick (2005) reports an extremely strong relationship of cognitive reflection scores with risk and time preferences. Rees-Jones and Taubinsky (2019) show that individuals fail to apply marginal tax rates and argue its relevance in designing tax schedules. Failure to properly condition on contingencies has been proposed as an explanation for the winner's curse (Charness and Levin, 2009). In a recent literature review on the topic, Niederle and Vespa (2023) connect failure of contingent thinking to college admission problems and health insurance choice.

[2]See Table 1 in Section 3.2 and Table C.1 in Appendix C.4 for a list of references for the ten tasks and a detailed description, respectively.

this and therefore dismiss contrasting actions as mistakes.

The experiment produces three main findings. First, for each task, there is a positive share of participants switching from their initial action. This shows that participants are willing to learn from others even when everyone faces the same underlying problem, and no one holds private information. Second, crucially, the impact of social learning on group outcomes differs across tasks. Overall, social learning has a significant negative impact on four tasks (RM, SSN, CN, and TM), a significant positive impact on four tasks (GF, CRT, EGB, and KS), and a non-significant impact on the remaining two tasks (AC and PC). The puzzling result that social learning amplifies errors for several cognitive biases can be explained in light of the conceptual framework. In fact, the correlation between relative confidence and relative performance has good predictive power on group gains from social learning. For tasks with a large, negative (positive) relative performance-relative confidence correlation, social learning has a negative (positive) impact on group performance, the group gains from social learning are generally increasing in the relative confidence-relative performance correlation. In other words, for tasks in which individuals' relative confidence is misaligned with relative performance, social learning leads to an amplification of errors caused by behavioral biases and, therefore, to a worsening of group outcomes.

In summary, this paper provides two key novel contributions. First, it documents that social learning can be detrimental to group outcomes, that is, it may increase the prevalence of a bias in a group of individuals. For example, reconsider the retail investors scenario. Depending on how well investors' relative confidence aligns with their ability to interpret information, social learning may either amplify or mitigate overreactions. When less accurate investors are relatively more confident, social learning may spread biased interpretations of new information; when more accurate investors are more confident, it may instead help correct them. Second, this paper identifies a mechanism, the alignment (or misalignment) of relative confidence and relative performance, that predicts when social learning helps or harms group decision-making. When the correlation between relative confidence and relative performance is negative, unbiased individuals observing suboptimal actions are more likely to switch to those actions than biased individuals observing optimal actions.

This paper contributes to several strands of the economics literature. First, it contributes to the experimental literature on social learning, but studies a different source of social informational value than the canonical framework. Much of the classic literature following Anderson and Holt (1997) considers environments in which individuals observe distinct private signals about an uncertain state and can learn from others because others possess information that they do not have (see, for example, Kübler and Weizsäcker (2004), Cipriani and Guarino (2005), Drehmann et al. (2005), Alevy et al. (2007), Eyster et al. (2018), Angrisani et al. (2021), and Conlon et al. (2022)). In contrast, in the environments studied here, participants face the same problem and no one has private information unavailable to others. Social learning is still potentially valuable because individuals may differ in how well they interpret or solve a common problem. The paper therefore studies social learning under internal uncertainty: whether and when individuals recognize that others may understand the same problem better than they do. The paper also relates to the theoretical literature on social learning with misspecified models, and in particular to Bohren and Hauser (2021), who characterize long-run beliefs when agents heterogeneously misinterpret the information content of others' actions. While their focus is on structural misspecification of the signal environment, the present paper studies a complementary source of distortion: miscalibration of relative confidence about the quality of observed answers.

The paper also differs from the canonical framework in the object of interest. Rather than studying whether social learning aggregates dispersed signals, I study whether it amplifies or mitigates errors induced by a broad set of economically relevant behavioral biases. In this sense, the paper is particularly related to Oprea and Yuksel (2021) and Grunewald et al. (2023), who also study how social interactions affect biased beliefs, and to Graeber et al. (2024), who study social learning about biased judgments in financial contexts. Relative to these papers, the focus here is on how the alignment between relative confidence and relative performance predicts the direction of social learning effects across tasks.

More broadly, the paper connects to work on learning in groups, observational learning, and decision-making from others' actions. A common theme in that literature is that observing others may improve choices when others possess different information, expertise, or experience (for a review, see Möbius and Rosenblat, 2014; see also Charness and Sutter, 2012, and Chamley, 2004). This has also been documented in strategic settings such as common value auctions, where observing others' actions and payoffs accelerates learning toward equilibrium (Armantier, 2004). The present paper isolates a complementary margin: even when individuals face the same information and the same problem, social learning may still improve or worsen outcomes depending on how accurately they infer the quality of others' decisions.

Second, this paper is related to the literature on overconfidence, and more specifically to the one on overplacement (following overconfidence classification by Moore and Healy, 2008). This literature has documented how individuals often erroneously believe to be better than others (Svenson, 1981; Camerer and Lovallo, 1999; Williams and Gilovich, 2008; Benoît et al., 2015), and how this belief varies with task difficulty, with an inversion of this tendency for harder tasks (Moore and Kim, 2004; Moore and Cain, 2007; Moore and Healy, 2008). While this literature focuses on the average overplacement and how this varies across different settings, in this work, I focus on the correlation between placement (relative confidence) and relative performance and its relation with the impact of social learning. Specifically, I show that when relative confidence is not well-calibrated, social learning amplifies the effect of cognitive biases.

Third, the paper contributes to the literature on internal uncertainty or imprecision. A central idea in this literature is that uncertainty need not be a structural feature of the environment, but may instead stem from the complexity of the reasoning process itself (Gabaix, 2019; Khaw et al., 2020; Enke and Graeber, 2023). The present paper shows that internal uncertainty is also central for social learning. Even when all individuals face the same information, social learning can be valuable because people are uncertain about whether they themselves or others have understood the problem better. To the best of my knowledge, this paper is also the first to elicit participants' assessments of the quality of others' answers in this type of setting, allowing me to study how relative confidence regulates learning behavior.

Finally, the paper connects to the literature on how behavioral biases aggregate into group and market outcomes (e.g. Russell and Thaler, 1985; Barberis and Thaler, 2003; Fehr and Tyran, 2005; Charness and Sutter, 2012; Lacetera et al., 2012). It is particularly closely related to Enke et al. (2023), who study whether institutions filter out behavioral biases and show that aggregation can either reduce or amplify mistakes depending on how well participants are calibrated about performance. The present paper studies a distinct institution of aggregation: social learning. This distinction matters because, unlike market or voting aggregation, social learning operates through decentralized imitation decisions. As a result, whether group outcomes improve depends not only on confidence in one's own answer, but on whether individuals assign relatively more confidence to better answers than to worse ones when they observe others. The paper's contribution is to show that this relative-confidence alignment predicts when social learning helps and when it harms.

The remainder of the paper is structured as follows. Section 2 develops a conceptual framework to convey the key intuitions and derive testable predictions. Section 3 details the experimental design and procedures. Section 4 presents results on social learning and relative confidence, Section 5 on the impact of social learning on group performance and its across-tasks heterogeneity, and Section 6 on the mechanisms behind such heterogeneity. Section 7 concludes and discusses limitations and potential future research avenues.

2 Conceptual Framework

This section develops a stylized framework whose purpose is to derive testable predictions and guide the empirical analysis, rather than to provide a general microfounded model of social learning. The framework takes as given a reduced-form relationship between relative confidence and the relative quality of the observed action, and asks what that relationship implies for the welfare effects of social learning.

Social learning improves group outcomes when agents are more willing to adopt the observed action precisely when it is better than their own. It harms group outcomes when confidence is miscalibrated in the opposite direction, so that imitation systematically flows away from the optimal action.

2.1 Setup

Consider a population of agents performing an identical task. There is a set of available actions A and a unique optimal action $a^* \in A$. Agent i initially chooses action a_i, and let

$$X_i = \mathbf{1}(a_i = a^*) \tag{1}$$

indicate whether that action is optimal. Since the welfare object of interest is whether the chosen action is optimal, the framework tracks the binary state X_i rather than the full distribution over suboptimal actions.

Let

$$\theta = \Pr(X_i = 1) \tag{2}$$

denote the current population share holding the optimal action. Equivalently, in a finite group interpretation, θ can be read as the expected pre-learning group optimality rate. After choosing a_i, agent i observes the action a_{-i} of one randomly drawn partner. Let

$$X_{-i} = \mathbf{1}(a_{-i} = a^*) \tag{3}$$

denote whether the observed action is optimal. Because the partner is drawn at random, $\Pr(X_{-i} = 1) = \theta$.

After observing a_{-i}, agent i can either keep their own action or switch to the observed one. Let θ_{SL} denote the post-learning group optimality rate. The gain from social learning is

$$\mathcal{G} = \theta_{SL} - \theta. \tag{4}$$

2.2 Relative Confidence and Social Learning

Agent i holds two confidence assessments: $c_{i,i} \in (0,1)$, the confidence that their own action is optimal, and $c_{i,-i} \in (0,1)$, the confidence that the observed action is optimal. The learning rule is

$$l_i = \begin{cases} a_{-i} & \text{with probability } \gamma\mu_i, \\ a_i & \text{with probability } 1 - \gamma\mu_i, \end{cases} \tag{5}$$

where $\gamma \in (0,1]$ captures how sensitive switching is to relative confidence, and

$$\mu_i = \frac{c_{i,-i}^2}{c_{i,-i}^2 + c_{i,i}^2}. \tag{6}$$

Hence the switching probability $\gamma\mu_i$ is strictly increasing in the relative-confidence ratio $c_{i,-i}/c_{i,i}$.

It is convenient to work with log relative confidence:

$$d_i \equiv \log\left(\frac{c_{i,-i}}{c_{i,i}}\right). \tag{7}$$

Under the switching rule above,

$$\mu_i = \Lambda(2d_i), \qquad \Lambda(z) \equiv \frac{e^z}{1+e^z}. \tag{8}$$

The key reduced-form assumption of the framework is that

$$d_i = \alpha + \beta(X_{-i} - X_i), \tag{9}$$

where $\alpha \in \mathbb{R}$ and $\beta \in \mathbb{R}$.[3] The parameter α captures a baseline tendency to favor the observed action relative to one's own. The parameter β, which I refer to as the *confidence–quality alignment parameter*, captures whether relative confidence moves in the right direction when the observed action is better or worse than one's own. When $\beta > 0$, the observed action receives more relative confidence when it is better than the agent's current action, and less relative confidence when it is worse. When $\beta < 0$, the opposite holds.

2.3 Predictions

Prediction 1. *The switching probability $\gamma\mu_i$ is strictly increasing in relative confidence $c_{i,-i}/c_{i,i}$.*

This follows directly from the learning rule and underpins the remaining predictions. It is tested empirically in Section 4.

Prediction 2. *If $\beta > 0$ ($\beta < 0$) and $\theta \in (0,1)$, then $\mathcal{G} > 0$ ($\mathcal{G} < 0$). Moreover, for any $\theta \in (0,1)$, $\mathcal{G}$ is strictly increasing in β.*

When $\beta > 0$, imitation flows in the right direction on average: agents who observe a better action are more likely to recognize it as such and adopt it, while agents who already hold the better action are less likely to abandon it. When $\beta < 0$, imitation flows in the wrong direction. A formal proof is provided in Appendix B.

Prediction 3. *If $\beta > 0$ ($\beta < 0$) and the initial distribution of actions is non-degenerate, then iterated social learning leads the group optimality rate to converge to 1 (to 0).*

Here non-degenerate means that the initial share of agents holding the optimal action lies strictly between zero and one. Under $\beta > 0$, each round of social learning shifts probability mass from suboptimal to optimal actions more strongly than in the reverse direction, so the optimal action diffuses through the population. Under $\beta < 0$, the same logic pushes the population toward a suboptimal consensus. A formal proof is provided in Appendix B.

[3] This log-ratio specification is adopted purely as a reduced-form normalization. It preserves positivity of confidence levels and yields closed-form predictions. The framework does not claim that agents literally compute log-odds; any strictly increasing specification with the same sign pattern would deliver the same qualitative implications.

3 Experimental Design

The aim is to design an experimental framework to investigate two main questions. First, does social learning reduce or amplify errors induced by biases? Second, does the impact depend on the type of bias? If yes, what are the mechanisms driving this difference?

To answer these questions, I set up an online experiment using ten different cognitive tasks, with the following features: (i) each task reflects a well-studied, economically relevant, cognitive bias; (ii) tasks have simple instructions and relatively short completion time; (iii) in each task there should be room for learning, that is it should not be too easy or too hard to learn from other participant's answers; (iv) tasks should feature no external uncertainty.[4] The order of the tasks is randomized.

3.1 Tasks Selection

The ten selected tasks are a subset of the fifteen tasks used in Enke et al. (2023), selected based on the fitness for the social learning framework and the absence of external uncertainty.[5] The absence of external uncertainty is relevant for two identification purposes. First, with the presence of uncertainty in the task, it is not possible to disentangle underreaction (overreaction) from underlearning (overlearning), despite the fact that these have possibly different root causes.[6] Second, and related, part of the goal of this work is to study the role of relative confidence in regulating social learning. The absence of uncertainty in the tasks allows for a cleaner identification of this effect. Hence, *Belief Updating* (BU) and *Base Rate Neglect* (BRN) tasks are excluded from the set of 15 tasks used in Enke et al. (2023). I also excluded 3 other tasks: *Iterated Reasoning* (IR), *Equilibrium Reasoning* (EQ), and *Wason Task* (WAS). The first two have been excluded because recognizing an optimal (or improving) answer would have been trivial for participants, making the learning process uninteresting to study. Relatedly, from pilot data, it emerged that 100% of participants switched in the learning phase of *Wason Task*, leading to no variability in one of the key outcomes of the experiment. Table 1 lists the ten selected tasks and the associated behavioral bias, and Table C.1 in Appendix C.4 provides a more detailed description. Screenshots of task instructions are provided in Appendix C.3.

[4] For additional details on how tasks have been selected, see Section 3.2. See Table C.1 in Appendix C.4 for a list and detailed description of the ten tasks.

[5] See their work for a discussion of how the tasks have been selected to fulfill the criterion of economic relevance.

[6] See Section 1 for a discussion on this point.

Task	Bias/Description
Information Processing and Statistical Reasoning	
Correlation neglect (CN)	Failing to account for non-independence of data in inference.
	Adaptation of tasks from Enke and Zimmermann (2019).
Gambler's fallacy (GF)	Failing to properly attribute independence to iid draws.
	Coin flipping task adapted from Dohmen et al. (2009).
Sample size neglect (SSN)	Failing to account for effect of sample size on precision of data.
	Adaptation of hospital problem from Kahneman and Tversky (1972); Bar-Hillel (1979).
Regression to mean (RM)	Failing to account for noise / failure to recognize regression to the mean.
	Adaptation of task from Kahneman and Tversky (1973).
Acquiring-a-company (AC)	Failing to properly condition on contingencies, à la the Winner's Curse.
	Bidding task against computer as in Charness and Levin (2009).
Logic	
Cognitive reflection test (CRT)	Following intuitive but misleading 'System 1' intuitions.
	Adaptation of Frederick (2005).
Constrained Optimization	
Knapsack (KS)	Failure to identify optimal bundle in constrained optimization problem.
	Knapsack problems taken from Murawski and Bossaerts (2016).
Financial Reasoning	
Thinking at the Margin (TM)	Thinking about average instead of marginal costs/benefits.
	Adaptation of marginal tax task from Rees-Jones and Taubinsky (2019).
Portfolio choice (PC)	Failure to construct efficient portfolios due to 1/N heuristic.
	Choose optimal portfolio vs. dominated 1/N portfolio.
Exponential growth bias (EGB)	Underestimate the exponential effects of compounding.
	Interest rate forecasting problem adapted from Levy and Tasoff (2016).

Table 1: Selected tasks descriptions and references, as reported in Enke et al. (2023).

3.2 Structure and Experimental Conditions

For each task, participants: (i) provide their answer for the current task; (ii) report their confidence about their answer; (iii) are shown an answer from another participant; (iv) provide an assessment of the probability of the observed answer being optimal; (v) participants are given a chance to change the answer provided in (i). This structure concerns the *Baseline* condition. Table 2 summarizes each treatment condition's key features and differences. In what follows, I provide additional details on the elicitation procedures in the *Baseline* condition. Afterward, I illustrate more in-depth the structure and the purpose of the additional treatments.

Treatment	Observe Others' Confidence	Observe Multiple Answers	Multiple Learning Rounds	Multiple Tasks
Baseline	✗	✗	✗	✓
OtherConf	✓	✗	✗	✓
GroupLearning	✓	✓	✓	✗

Table 2: Experimental Conditions Main Features

Confidence Elicitation

Once participants provide their solution to the task, they are asked to provide their confidence level. This elicitation takes place for all participants, in each task. The question is posed in terms of certainty about decision optimality, following Enke and Graeber (2023) and Enke et al. (2023). Figure 1 shows a screenshot of confidence elicitation.

As argued by Enke et al. (2023), confidence elicitation may impact the following decisions, which may speak against a within-subjects design, such as the one employed in this paper. On the other hand, a within-subject design allows for establishing a more direct link between confidence elicitations and social learning behavior, which of primary relevance to study the mechanism illustrated in Section 6.

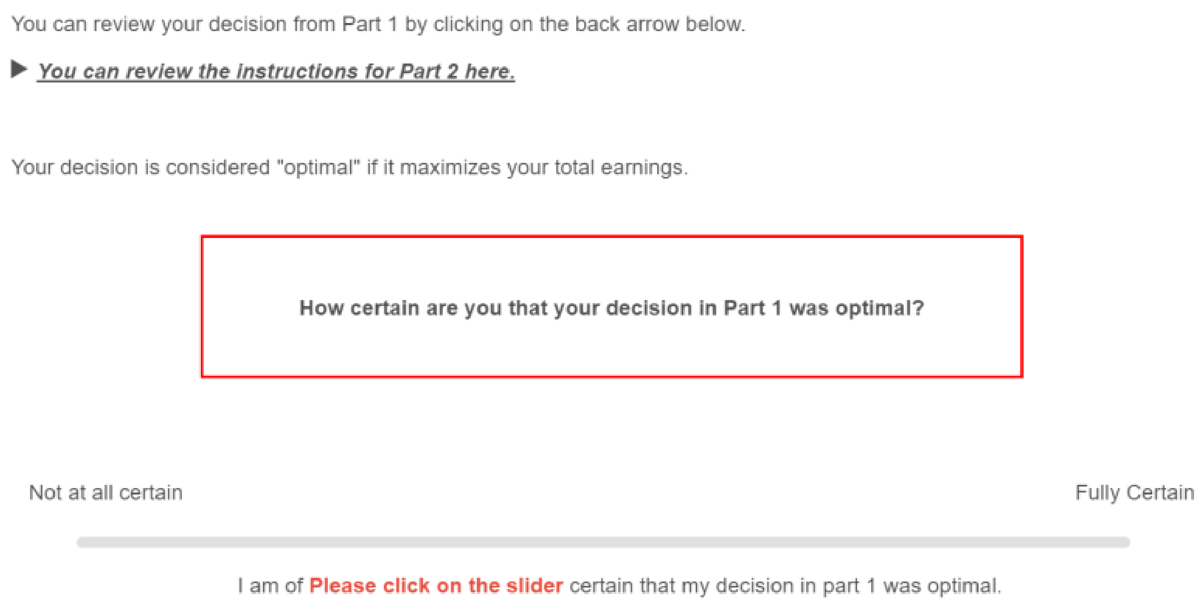


Figure 1: Example of $c_{i,i}$ elicitation.

Learning Phase

In the learning phase, participants observe an answer provided by another participant to the exact same task (a_{-i}). After that, they provide their assessment of the probability that the observed answer is optimal ($c_{i,-i}$) and, finally, participants may change their initial answer to a new *learning answer* (l_i). Importantly, a_{-i} is *always different* from a_i. This design choice maximizes statistical power, as observations where $a_i = a_{-i}$ would be uninformative about social learning behavior and excluded from the analysis. It also ensures that every participant faces a genuine learning opportunity, making the within-task variation in relative confidence and switching behavior directly comparable across participants.[7] This could raise three kinds of concerns. First, selecting which answer to show to participants based on their previous actions could generate an endogeneity problem. In short, I tackle this by applying a correction to the measure of net aggregate gains from social learning. Details on this are provided in Section 5. As

[7]Clearly, this statement relies on the assumption that a participant would stick with their initial choice, after observing an identical action.

a further robustness check, I conduct an additional preregistered study (AsPredicted #282144) in which participants are shown a randomly drawn answer from another participant, removing the contrasting-answer restriction entirely. The results of this study are discussed in Section 6. Second, one may be worried that participants are being deceived, as they are not being shown a random answer. However, as reported in Figure C.4 in Appendix C.1, the instructions of the task state that they will be shown another participant's answer, without specifying that the answer is randomly drawn. Figure 2 and Figure 3 show a screenshot of $c_{i,-i}$ elicitation and of l_i elicitation respectively. Third, participants' answers may change if they believe that the answers that are being shown are non-random or computer generated. To tackle this issue, in the final block of the experiment, participants are asked to report if they had any comments on the shown answers from other participants. This way, it is possible to exclude participants who report concerns about the shown answers' legitimacy.

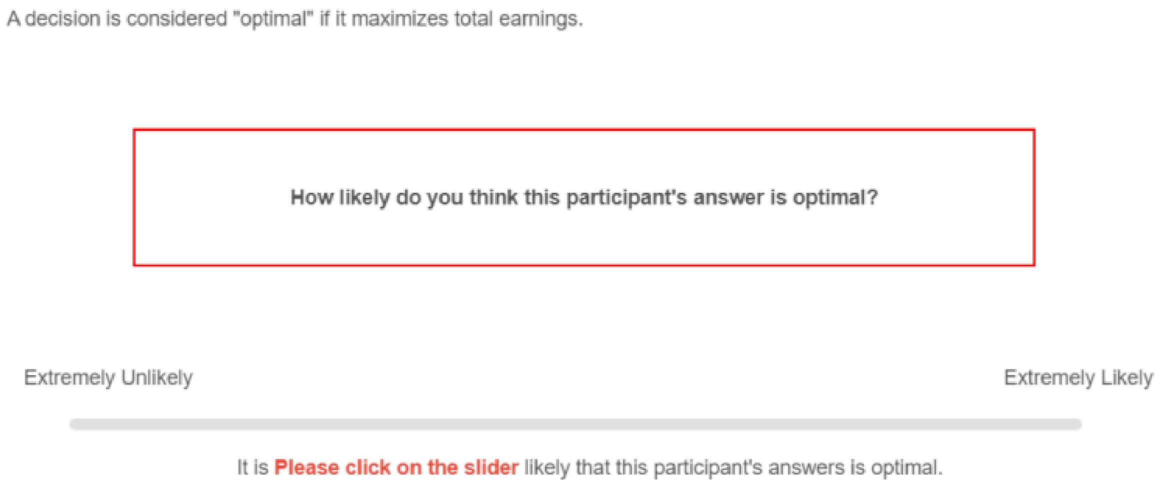


Figure 2: Example of elicitation of observed answer optimality ($c_{i,-i}$).

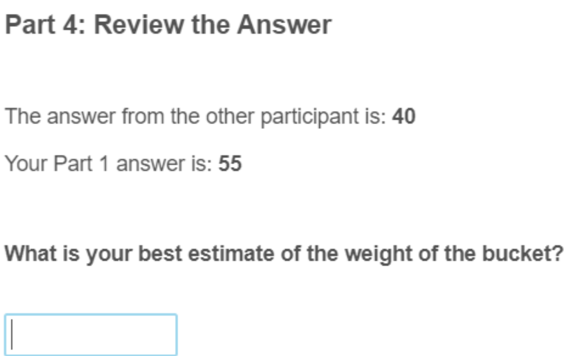


Figure 3: Example of learning action (l_i) elicitation from Correlation Neglect (CN) task.

OtherConf **Condition**

In the *OtherConf* treatment, participants observe directly other participants' confidence, as opposed to guessing the probability of the observed answer being optimal. Specifically, given the observed answer (a_{-i}), participants are shown the median confidence level associated with that specific answer.[8]

In principle, there may be social learning settings in which individuals infer (e.g. social learning settings in which only actions or performances from others are available, à la Moore and Healy, 2008) or observe (e.g. in a conversation or a debate) others' confidence levels. Relatedly, it has been shown that inferred or observed levels of confidence influence the extent to which individuals react to information provided by others (e.g. Van Zant and Berger, 2019; Amelio, 2022), and also that individuals seem to be sophisticated and strategically manipulate their confidence level to be more persuasive (Schwardmann and Van der Weele, 2019). Hence, this treatment is a natural extension to the *Baseline* treatment. The aim is to assess whether and to what extent the results in the *Baseline* treatment extend to a setting in which participants observe others' confidence, which is a natural and relevant social learning framework per se.

GroupLearning **Condition**

Three additional questions that naturally arise starting from the *Baseline* condition are: (i) If social learning takes place in groups of multiple participants, as opposed to pairs, how does this impact findings? (ii) Does having multiple learning rounds, as opposed to one, improve or hinder the impact of learning on group outcomes? (iii) Do people converge on a specific, possibly incorrect, answer? This condition tackles all of these questions.

In the *GroupLearning* condition, participants first independently complete a task.[9] As in other treatments, they provide an answer and subsequently their confidence level. Afterward, each participant is matched with three other participants, forming a group of four. The groups are formed to contain two participants who answered optimally and two who did not.[10] All participants observe the answers and confidence levels of all group members. Finally, each participant may change their answer and their confidence level in their (potentially different) answer. This procedure is repeated for a total of three rounds, in which the four participants stay unchanged. A focus on the first learning round allows to investigate the impact of social learning in a group, as opposed to social learning from an individual, on group outcomes. Analyzing the answers' dynamics over rounds, with a special focus on the last one, allows to investigate whether repeated social learning leads to a different impact on group outcomes compared to single-round learning. Additionally, it is possible to look for evidence of convergence on a specific answer and assess if this convergence is beneficial or detrimental to the group.

[8] An alternative design choice could have been to show the confidence level of a random participant who provided a specific answer. However, this approach would have generated noisier data.

[9] Unlike the other two experimental conditions, participants only solve one task. Out of the ten tasks in the *Baseline*, two have been selected: CRT and RM (see the following section for more details on each task).

[10] To not to distort participants' perception about the distribution of answers and to minimize attrition rates, two tasks with an optimality rate as close as possible to 50% have been selected.

3.3 Logistics

The experiment was conducted on Qualtrics, using Prolific as a recruiting platform. In total, 1700 participants were recruited, of which 300 for *Baseline*, 200 for *OtherConf*, and 1200 for *GroupLearning*. For the *Baseline* treatment, participants took on average approximately 22 minutes to complete the study and received on average £3. Hence, the hourly wage was approximately £8. The median completion time for *OtherConf* was approximately 20 minutes with a comparable compensation to *Baseline*. Finally, for *GroupLearning* the median time was approximately 5 minutes, including the time to be matched with other participants, with an average compensation of approximately £0.8.

Following instructions, participants had to complete a set of comprehension questions, to ensure that they successfully understood the essential parts of the experiment. Participants who failed to answer at least one of the questions were screened out. Appendix C.2 reports screenshots of all the comprehension questions. Both sample sizes and exclusion restrictions have been preregistered.

4 Learning Behavior and Relative Confidence

The first piece of evidence I present concerns general patterns in social learning behavior and how this behavior is modulated by relative confidence. Figure 4 reports the probability of switching by task, that is the share of participants such that $a_i \neq l_i$.

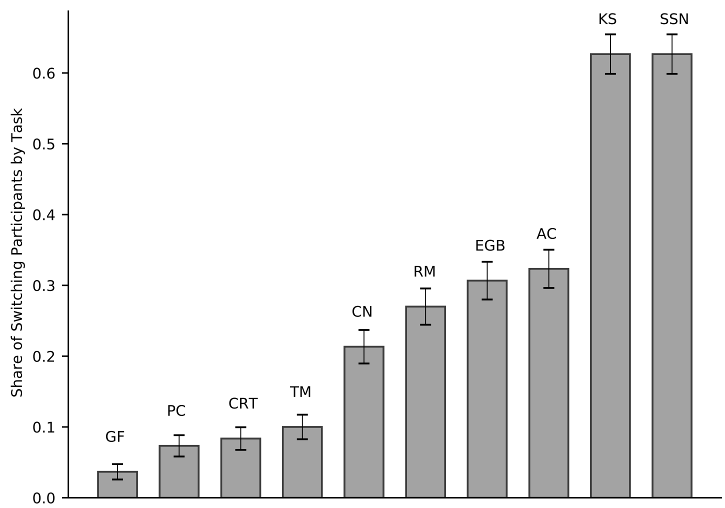


Figure 4: Switching rates by task. A participant is classified as a switcher if their learning action (l_i) is different from their initial action (a_i). Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF=Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking.

Two aspects are worth stressing. First, there is consistent heterogeneity in switching rates across tasks. Second, switching rates are economically meaningful and positive in every task, indicating that participants are willing to learn from others even in the absence of external uncertainty.[11]

Given that participants are willing to switch to other actions, what regulates their learning behavior? Figure 5 reports the share of participants with $c_{i,i} < c_{i,-i}$ for switchers and non-switchers respectively, that is the probability of being relatively less confident in a_i compared to a_{-i}, conditional on having switched or not. Switchers in each task always exhibit a significantly higher rate of participants with $c_{i,i} < c_{i,-i}$, although this share varies substantially across tasks.

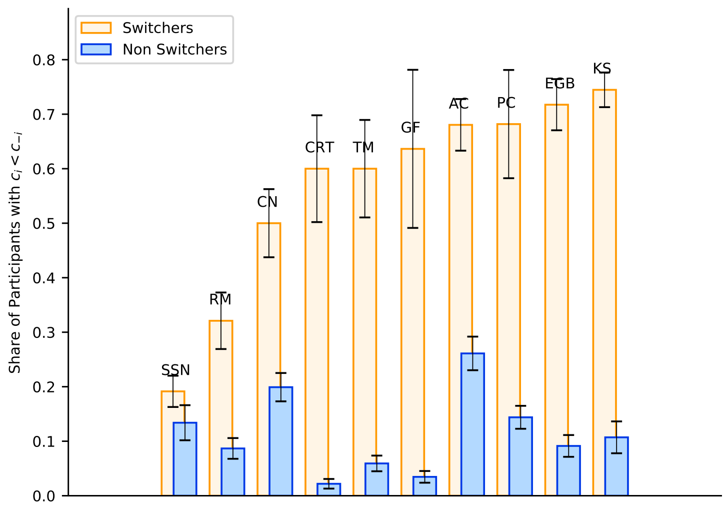


Figure 5: Share of participants with reported confidence lower than the assessed probability of a_{-i} being optimal, by task. Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF=Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking.

Figure 6 reports additional evidence on the relationship between relative confidence and switching behavior, showing the PDF of relative confidence for switchers and non-switchers. Relative confidence is computed as the difference between confidence and assessed probability of a_{-i} being optimal.[12] The two distributions are significantly different,[13] with the non-switchers dis-

[11]Clearly, the propensity to switch from the initial answer will also depend on task difficulty. However, as shown by Enke et al. (2023), performance and confidence are not necessarily positively correlated, and in some tasks, confidence assessments may be systematically miscalibrated. This calibration, combined with the accuracy with which participants can assess the observed answer quality, determines the mediating effect of relative confidence on the impact of social learning on group outcomes, as argued more in-depth in Section 6.

[12]The reason why relative confidence is not defined as the ratio between $c_{i,i}$ and $c_{i,-i}$, as in Section 2, is to avoid discarding observations in which $c_{i,i} = c_{i,-i} = 0$. All results are robust to the alternative definition.

[13]A t-test comparing the mean relative confidence for switchers and non-switchers rejects the null with $p < 0.001$. A two-sample K-S test rejects the null that the two distributions are drawn from the same probability distribution with $p < 0.001$.

tribution being more right-skewed, confirming that relative confidence is an important driver of social learning behavior.

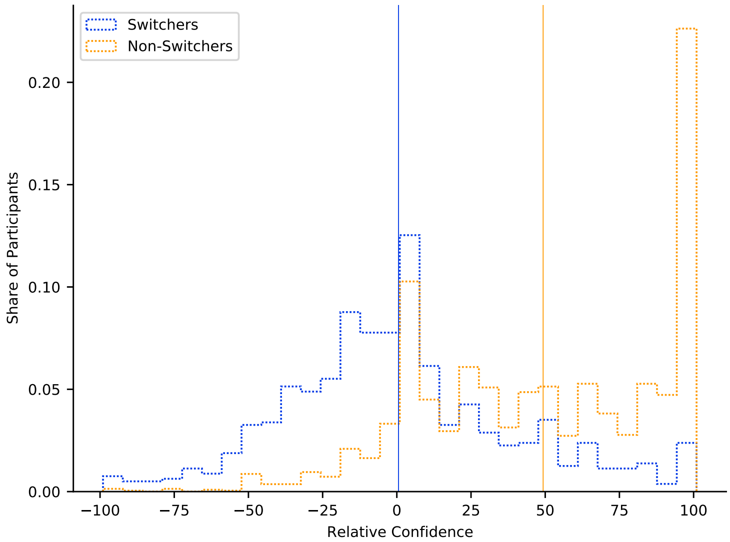


Figure 6: PDF of relative confidence, comparing switchers and non-switchers. Relative confidence is constructed as $c_{i,i} - c_{i,-i}$. The vertical lines represent the distribution means.

Table 3 formalises this relationship, reporting the results of a regression of switching behavior on relative confidence. The coefficient on relative confidence is positive and highly significant across all specifications, and is stable when controlling for trial order and participant demographics, confirming that relative confidence is a primary driver of switching behavior.

	(1)	(2)	(3)	(4)
	Switch			
Relative confidence	0.449***	0.449***	0.455***	0.413***
	(0.070)	(0.069)	(0.071)	(0.063)
Constant	0.425***	0.438***	0.454***	0.412***
	(0.072)	(0.073)	(0.074)	(0.023)
Trial order	No	Yes	Yes	Yes
Demographics	No	No	Yes	Yes
Task FE	No	No	No	Yes
Observations	2,999	2,999	2,999	2,999
Clusters (task)	10	10	10	10
R^2	0.219	0.219	0.221	0.361

Standard errors adjusted for two-way clustering at the task and participant level in parentheses. Dependent variable is a dummy equal to 1 if the participant switched from their initial answer. Relative confidence is constructed as $c_{i,i} - c_{i,-i}$, normalised to the range $[-1, 1]$. Demographics include gender, age, and education level. The VCV matrix in columns (3) and (4) was non-positive semi-definite; the adjustment from Cameron et al. (2011) was applied.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3: Regression of Switching Behavior on Relative Confidence

5 Impact of Social Learning

Given that participants are willing to switch from their initial action, does learning affect positively or negatively group performance? To answer this question I compare the optimality rates in each task before and after the learning phase. The optimality rate pre-learning (post-learning) is the share of participants choosing the optimal action before (after) learning.

Participants may fall into one of the following four categories, depending on their pre-learning response and their learning behavior: (i) overlearner, (ii) optimal switcher, (iii) underlearner, and (iv) optimal non-switcher. Table 4 summarises the features of each category. Note that, in measuring the impact of social learning, only overlearners and optimal switchers matter, with the former being associated with losses and the latter with gains. Underlearners also represent a sub-optimal behavior, however, by construction, they do not impact changes in optimality rates at the group level. At the same time, the share of undelearners is interesting, as it represents an upper bound for gains from social learning.[14] Figures A.3 and A.2 in the Appendix report the share of overlearners and optimal switchers, respectively, by task. These

[14]Moreover, undelearners represent a consistent share of participants across tasks, approximately 39%, that is more than half of the participants who do not switch at all.

figures are descriptive only: because participants are always shown an answer different from their own, the raw shares are mechanically affected by the pre-learning optimality rate and therefore are not, by themselves, an appropriate object for comparing the impact of social learning across tasks.

	Switcher	**Non-Switcher**
Optimal a_i	*Overlearner*	*Optimal non-switcher*
Non-optimal a_i	*Optimal switcher*	*Underlearner*

Table 4: The four possible categories of a participant in the learning phase. Optimal/non-optimal a_i refers to the optimality of the action chosen in the pre-learning phase. Switcher/non-switcher refers to the participant sticking or not to their pre-learning action.

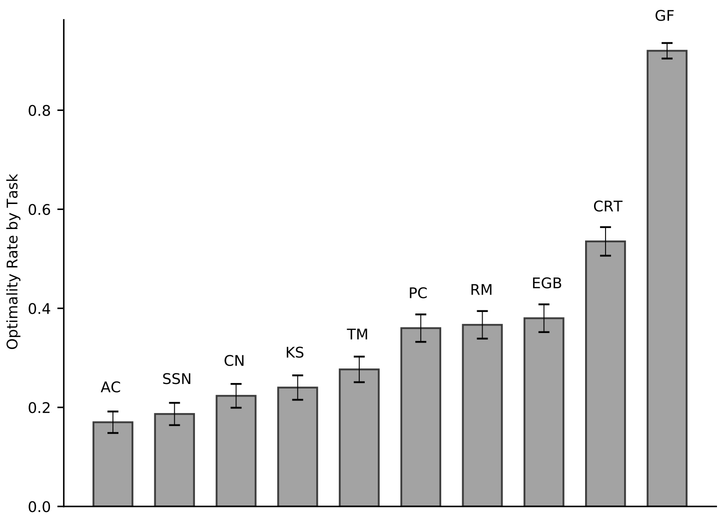


Figure 7: Share of participants choosing the optimal action, before learning, by task. Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF= Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking.

Figure 7 reports the pre-learning optimality rates by task, showing the heterogeneity in performance across tasks. This heterogeneity is particularly relevant given the design choice discussed in Section 3: in the learning phase, participants are always shown a different answer from the one they initially chose. More specifically, participants who took the optimal action were shown a non-optimal answer, and, vice versa, participants who took a sub-optimal action were shown the optimal answer.[15] Hence, the extent to which there is room for gains or losses from learning depends on the pre-learning optimality rate. For example, in a task with a quite low optimality

[15] The rationale behind this design choice is illustrated in Section 3.

rate, most participants were shown the optimal answer, implying, ceteris paribus, a higher probability of a gain from learning. For this reason, simply taking the difference in optimality rates by tasks would not be a clean measure of group gains from learning. To tackle this issue, I build a weighted measure of group gains from learning, taking into account the actual optimality rate in each task:

$$w\_group\_gains_k = p_k \cdot group\_gains_k - (1 - p_k) \cdot group\_losses_k,$$

where p_k is the pre-learning optimality rate in task k; $group\_gains_k$ is the share of participants who switched to an optimal answer from an incorrect one, in task k; vice versa, $group\_losses_k$ is the share of participants who switched from an optimal answer to an incorrect one, in task k. Referring to Table 4, this measure is comparing the *proportions* of *overlearners* and *optimal switchers*. In other words, the weighted group gains are a weighted sum of gains and losses from learning. The weights represent, respectively, the probability of being exposed to an optimal action (p_k) and hence having the chance to gain from learning, and the probability of being exposed to a sub-optimal action $(1 - p_k)$ and incurring the possibility of a loss from learning. Alternatively, the weighted gains (losses) can be interpreted as the probability of switching to a correct (wrong) action, having answered wrongly (correctly) in the first step. [16]

[16]This interpretation holds under the assumption that individuals observing an action identical to their initial choice would not switch to another action. Note that $group\_gains_k$ and $group\_losses_k$ are defined as shares of the full sample rather than as conditional rates within subgroups. Specifically, $group\_gains_k = (1 - p_k) \cdot \Theta_k^{not}$ and $group\_losses_k = p_k \cdot (1 - \Theta_k^{opt})$, where Θ_k^{not} is the post-learning optimality rate among initially incorrect participants and Θ_k^{opt} is the post-learning optimality rate among initially correct participants. Under this definition, the weighted measure simplifies to $p_k(1-p_k)(\Theta_k^{not} - (1-\Theta_k^{opt}))$, which coincides with the counterfactual expected net gain under random answer assignment, where an initially incorrect participant would see the correct answer with probability p_k and an initially correct participant would see the wrong answer with probability $1 - p_k$.

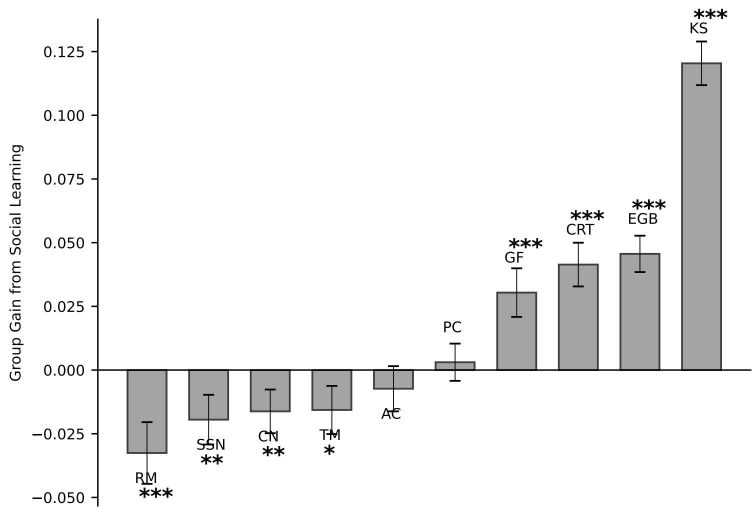


Figure 8: Group gains from social learning, by task. The weighted net gain measure, for each task, is built by weighting gains and losses with the optimality rate in and its complement on 1, respectively. Error bars represent standard errors. Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF= Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking. The stars indicate the following p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Figure 8 shows clearly that the effect of social learning can be both beneficial and detrimental for group outcomes. Over the ten tasks, four exhibit a significant loss, four a significant gain, and two no significant impact.[17] The net gains range from a loss of approximately 3% (for the RM task) to a gain of approximately 12% (for the KS task). It is interesting to note that these net variations are almost always the result of both gains and losses from social learning, which are usually of a larger, and quite relevant, magnitude.[18] Figures A.2 and A.3 report unweighted gains and losses respectively. The results are very similar for the *OtherConf* condition (see Figure A.9 in Appendix A.7). These comparisons should nonetheless be interpreted with some caution, since *OtherConf* differs from *Baseline* not only in the information shown to participants, but also in that the confidence associated with the observed answer is directly provided rather than inferred by the participant. Interestingly, group gains from learning differ depending on participants' gender, with females gaining less from learning on average (see figure A.8 in Appendix A.6).

In what follows I show how results of *Baseline* and *OtherConf* conditions are robust when social learning takes place in groups and iteratively. Afterward, since the impact of social learning differs across tasks, the following section explores the mechanisms behind this heterogeneity. More

[17] Applying Benjamini-Hochberg correction for multiple comparisons across the ten tasks, the significant results for SSN, CN, and TM are not robust to the correction, while all other significant results are unchanged.

[18] For example, the RM task features 7% of participants switching to the optimal answer and 10% of participants switching from the optimal answer to an incorrect answer.

specifically, it shows how the relative confidence-relative performance correlation is predictive of group gains from social learning.

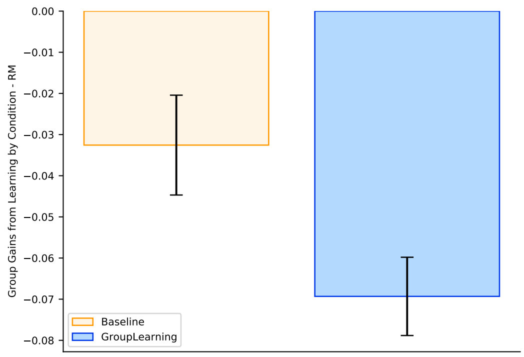
Group Learning and Multiple Learning Rounds

All the results shown so far concern learning from a single action. In other words, participants were observing another individual participant's answer and deciding whether to stick to their initial action or switch to a different one. However, are results robust to social learning taking place with multiple individuals at the same time? In the *GroupLearning* condition, I focus on this question, using two of the ten tasks studied in the other conditions, RM and CRT. The first (second) has been selected to study group learning in the case of a task characterized by a negative (positive) effect of social learning on group outcomes.[19] Figures 9a and 9b compare, for the RM and the CRT task respectively, the group gains from learning for the *Baseline* condition and the *GroupLearning* condition. For the latter, it is important to specify that the gains are calculated for the first round of learning.[20,21] This allows us to explore the question of whether the results, in terms of the negative impact of social learning on group outcomes, are robust to learning taking place in groups. In fact, the figure shows that not only the result is robust, but that for the *GroupLearning* condition, the losses from learning are approximately doubled for the RM task and the gains approximately tripled for the CRT task.

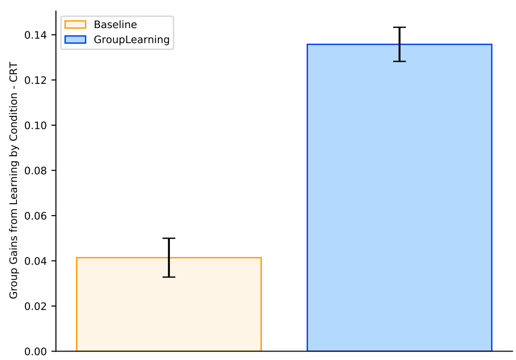
[19]For additional details on the structure of the condition and on the selection criteria see Section 3.2. Broadly, the two tasks were selected using the optimality rate before learning. The aim was for the two tasks to have an optimality rate as close as possible to 50%, since this is how groups are built in this treatment. This criterion is preregistered.

[20]This comparison should nonetheless be interpreted with some caution, as the *GroupLearning* condition differs from *Baseline* not only in the number of observed answers and learning rounds, but also in that participants know their confidence reports are visible to others, which may affect confidence reporting behavior.

[21]The gains are calculated using the weighted measure illustrated in Section 5, although the share of participants with the optimal answer is, by construction, 50%. Hence, the only effect the weighting has in this case is of halving the actual difference between optimal and sub-optimal switchers. Using the latter directly would make the two measures not comparable.

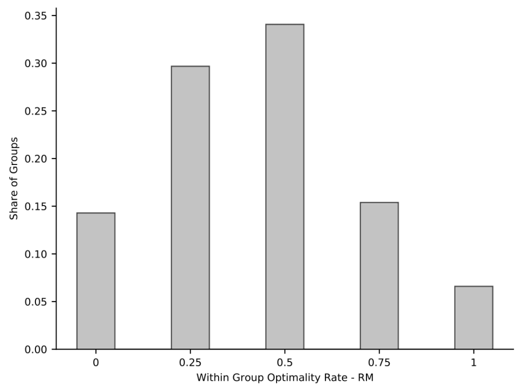


(a) Group gains (losses) from social learning, for the RM task, comparing the Baseline and the GroupLearning conditions. For the latter, the gains are calculated for the first round of learning. Error bars represent standard errors.

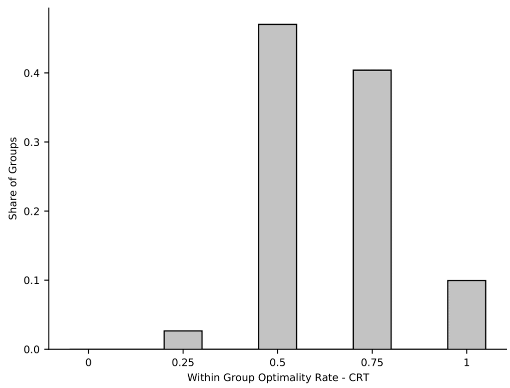


(b) Group gains (losses) from social learning, for the CRT task, comparing the Baseline and the GroupLearning conditions. For the latter, the gains are calculated for the first round of learning. Error bars represent standard errors.

This condition introduces another addition to the *Baseline*, that is participants in a group engage in two additional rounds of learning after the first. Figures 10a and 10b report the second key result of this section, concerning the impact of iteration, that is of multiple learning rounds. The figure shows the distribution of within-group optimality rates in the last learning round for both RM and CRT. The results for RM show that approximately 15% of the groups end up with all group members choosing the sub-optimal action and that 30% of the four-participants groups conclude the learning rounds with only one member having stuck with the optimal action. In line with the results shown in Figures 9b and 8, the results for the CRT task are even stronger. Figure 10b shows how no group has zero optimal answers at the end of the learning rounds and that slightly more than 2% of the groups have only one member sticking to the optimal answer. Hence, for CRT, social learning iterations seem to be extremely effective, as almost no group is worse off than before learning.[22]



(a) Distribution of share of optimal answers within each group of four participants in the last learning round, for the RM task.



(b) Distribution of share of optimal answers within each group of four participants in the last learning round, for the CRT task.

[22]Recall that, by construction, the initial distribution is that all groups have 50% optimality rate.

Two additional observations are worth mentioning. First, for RM (CRT), the distribution is strongly skewed to the left (right). This means that most of the participants do not benefit (do benefit) from social learning, even when it takes place in groups and even when it is iterated over multiple learning rounds. Second, in line with Prediction 3, Figure 10a (10b) provides evidence for convergence towards the sub-optimal (optimal) action. By design, all groups start with a 50% optimality rate. Out of the groups that end up with a different optimality rate (approximately 65%), approximately two-thirds exhibit a lower one, that is more group members pick the sub-optimal action. A similar, but stronger, consideration can be made for CRT in which, as mentioned, almost all groups end up with a better optimality rate than the starting one. These observations support the view that the negative (positive) impact of social learning on group outcomes in the *Baseline* condition is robust to a setting in which: (i) multiple participants can observe each others' actions, and (ii) participants may learn from each other multiple times. This reinforces the result that the negative or positive impact of social learning documented so far is strongly related to the task or bias at hand. In the next section, I investigate this very aspect and show how a task-specific feature, the relative confidence-relative performance correlation, can account for how social learning impact differs across different tasks.

6 Relative Confidence-Relative Performance Correlation and Social Learning Impact

The previous section shows how the effect of social learning on group outcomes can vary for different behavioral biases. This section explores the mechanism proposed in the formal framework, based on relative confidence-relative performance correlation. Relative confidence here is defined as the difference between the participant's reported confidence in their answer ($c_{i,i}$) and their assessment of the probability of a_{-i} being optimal ($c_{i,-i}$). In other words, the relative confidence measure is $r_i = c_{i,i} - c_{i,-i}$.[23] Relative performance is a dummy variable, taking the value of 1 if the participant's action (a_i) is optimal and the action she is observing (a_{-i}) is not, and being equal to 0 if the opposite holds. Note that, given how a_{-i} is chosen in the experiment, there is a perfect correspondence between this relative performance dummy and a dummy for optimality. This is because participants whose a_i is optimal are always shown a sub-optimal a_{-i} and vice versa.

Figure A.1 shows how β varies across different tasks. This correlation can be interpreted as the average precision of participants' relative confidence assessments for a given task. For example, if $\beta < 0$, then, on average, participants who are observing the optimal answer (as their pre-learning answer was sub-optimal) will be more confident in their answer and, similarly, participants whose

[23]Encoding relative confidence this way allows to also account for the intensive margin of relative confidence in building our measures of interest. Results are robust to a dichotomic encoding of relative confidence, with the variable taking the value of 1 if $c_{i,i} > c_{i,-i}$ and 0 otherwise. Figure A.10 reports results equivalent to Figure 11 with this different encoding for relative confidence.

original answer is optimal, and are hence observing a sub-optimal answer, will be relatively less confident in their own answer.

The hypothesis is that this correlation translates into how social learning impacts group performance, through the way relative confidence modulates learning behavior. Table 5 provides formal evidence for this relationship, reporting the results of a regression of individual-level weighted net gains from social learning on the within-task relative confidence-relative performance correlation β. The coefficient on β is positive and highly significant across all specifications, and is stable when controlling for trial order and participant demographics, indicating that the relationship is not driven by ordering effects or sample composition. Figure 11 illustrates the same relationship at the task level, showing that the tasks with the largest (smallest) β also exhibit the largest gains (losses) from social learning.[24] This relationship does not seem to be predicted by task difficulty (see Figure 13b) nor by the type of elicited answer (comparing continuous and discrete tasks, see Figure 13a). Additionally, the results do not vary when considering only a sub-sample of participants who did not express any concern or doubt regarding the authenticity of the observed actions, as shown in Figure A.12.[25] Finally, this same result holds for the *OtherConf* condition, that is in the case of participants observing others' reported confidence levels, as shown in Figure 12.

[24] The confidence-performance correlation, or *confidence calibration*, used in Enke et al. (2023), is not predictive of gains from learning as the relative confidence-relative performance correlation, as reported in Figure A.11. This shows how confidence calibration is not sufficient to account for the impact of learning.

[25] As participants are always shown answers that are different from their initial choice, they may doubt the fact that the observed answers are genuine or human-generated. For this reason, at the end of the experiment participants are asked to answer an open-ended question, concerning doubts or comments they may have on observed answers. The figures in the Appendix report the two main results excluding participants who expressed doubts about the authenticity of the observed answers.

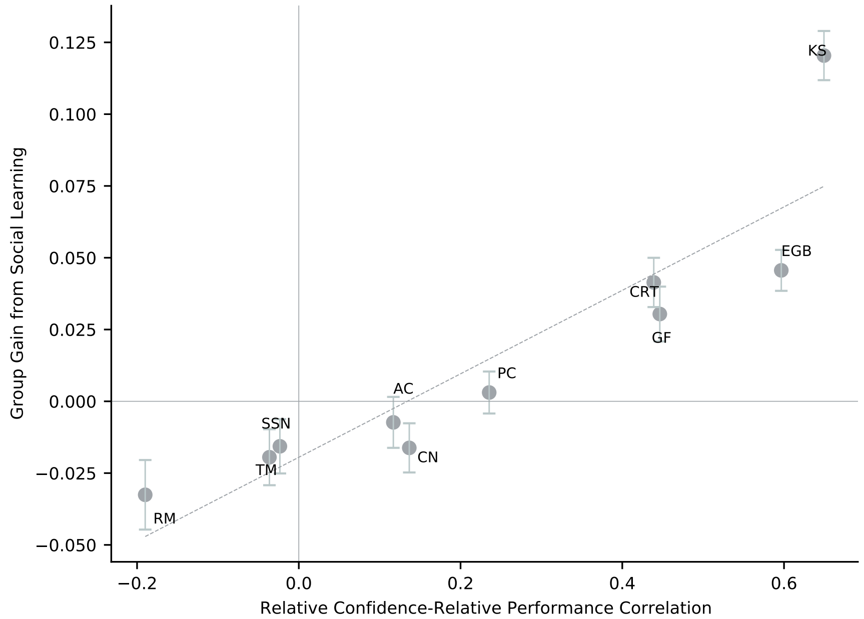


Figure 11: Scatter plot of relative confidence-relative performance correlation (on the x-axis) and the weighted group gain from social learning (on the y-axis). The error bars represent standard errors. Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF=Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking.

	(1)	(2)	(3)
	Group Gains from Social Learning		
β	0.145***	0.145***	0.145***
	(0.031)	(0.031)	(0.031)
Constant	−0.019***	−0.020***	−0.031
	(0.005)	(0.007)	(0.019)
Trial order	No	Yes	Yes
Demographics	No	No	Yes
Observations	2,999	2,999	2,999
R^2	0.058	0.058	0.061

Standard errors adjusted for two-way clustering at the task and participant level in parentheses. Dependent variable is the individual-level weighted net gain from social learning. β is the within-task relative confidence-relative performance correlation, estimated from the full sample and fixed within task. Demographics include gender, age, and education level. The VCV matrix in column (3) was non-positive semi-definite; the adjustment from Cameron et al. (2011) was applied.
$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Group Gains from Social Learning and β

Comparing the results in Figure 11 (*Baseline* condition) and Figure 12 (*OtherConf* condition) two main differences emerge. First, CN (correlation neglect) and TM (thinking at the margin) fall in different quadrants of the plane in each condition. Most interestingly, in *OtherConf*, CN exhibits a negative relative confidence-relative performance correlation and, in line with the framework, a group loss from learning, while in *Baseline* CN exhibits a group gain from learning. A potential explanation for this difference is that, for *Baseline*, CN is the task in which switching behavior is the most inconsistent with relative confidence. The latter is, on average, close to zero, indicating a high level of uncertainty from participants. However, in *OtherConf*, participants observe the confidence level associated with the observed action and seem to take it at face value. This may reduce the inconsistency between relative confidence and switching behavior, generating this difference between the two conditions. Second, comparing the intersection of the best-fitting line for the two conditions, the one in *OtherConf* is very close to zero, while the one in *Baseline* is strongly negative. The latter implies that, in the *Baseline* condition, for tasks with a zero relative confidence-relative performance correlation, there would be losses from learning. However, in relation to the previous point, this difference is mainly driven by the CN task, so it should be interpreted with caution.

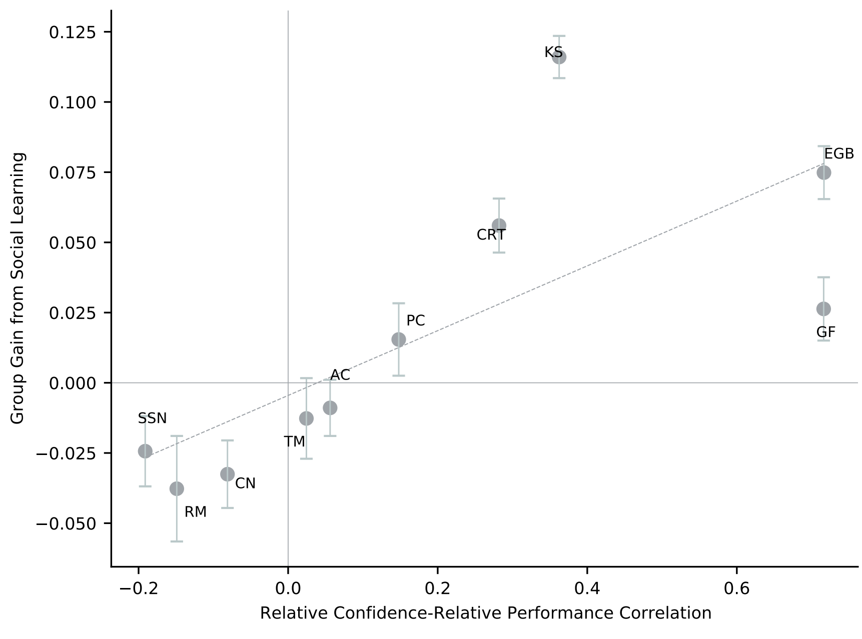


Figure 12: Scatter plot of relative confidence-relative performance correlation (on the x-axis) and the weighted group gain from social learning (on the y-axis), for the OtherConf condition. The error bars represent standard errors. Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF=Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking.

To further probe the robustness of this relationship and assess whether it holds beyond the specific design features of the *Baseline* condition, I conduct an additional preregistered study in which participants are shown a randomly drawn answer from another participant rather than a contrasting one.

Robustness to Random Answer Assignment

A potential concern with the *Baseline* design is that always showing participants a contrasting answer may artificially influence the relationship between relative confidence and group gains from social learning. To address this, I conduct an additional preregistered study (AsPredicted #282144) in which participants are shown a randomly drawn answer from another participant, removing the contrasting-answer restriction. The study covers five tasks — GF, SSN, RM, PC, and TM — selected from the original ten on the basis of having a discrete answer space, which makes random assignment unambiguous.[26] The β estimates from the original study are used as out-of-sample predictors of the direction of social learning effects in this new dataset, following the preregistration.

The primary test is whether the sign of the net gain from social learning in the new study matches the sign of β from the original study, for each of the five tasks. Under the null hypothesis of no predictive relationship, the expected number of correct signs is 2.5 out of 5. The results show that all five tasks exhibit the correct sign ($p = 0.031$, binomial test). A Spearman rank correlation between original β estimates and new net gains yields $\rho = 0.900$ ($p = 0.037$).[27] This means that the five tasks rank in the same order in terms of social learning outcomes in the new study as predicted by their β estimates from the original study. Figure A.13 in the Appendix reproduces Figure 11, adding the net gains from the new study as a separate set of points, with β values taken from the original study. The two sets of points trace a similar upward-sloping pattern, indicating that the relationship between β and social learning outcomes is not an artefact of the contrasting-answer design and holds out of sample under a cleaner identification strategy.

6.1 Relative Confidence Assessment and Task Features

A question that naturally arises from these results is: which features of a task determine the sign and magnitude of the relative confidence-relative performance correlation? A useful framing is that β reflects two components simultaneously: how well participants assess the quality of their own answer before observing others, and how well they assess the quality of the observed answer once it is shown. An explanation for cross-task differences in β must account for both components. As discussed below, explanations that focus only on the pre-learning phase (such

[26] Tasks with continuous answer spaces are excluded because the large dispersion of answers implies that a randomly drawn answer is unlikely to be either the correct answer or meaningfully different from the participant's own answer. This would result in a high share of uninformative learning opportunities, which is undesirable for a clean comparison with the original study. Discrete tasks suffer less from this problem, as the answer space includes either two or three options.

[27] Both results are identical across the full sample and the sample restricted to observations where the shown answer differs from the participant's own answer.

as task difficulty or the attractiveness of the wrong answer) fall short precisely because they disregard what happens during the learning phase.

Misleading Intuitions

A natural candidate explanation draws on the concept of misleading intuitions. Enke et al. (2023) operationalize this as the "peakedness" of the distribution of wrong answers within each task, defined as the frequency of the modal suboptimal answer, with higher peakedness indicating a more attractive incorrect answer. Figures A.4, A.5, A.6 and A.7 show that CRT, EGB, RM, and SSN are all characterized by high peakedness. However, the first two exhibit a large positive β while the latter two exhibit a negative one, indicating that peakedness alone does not predict the sign of β.

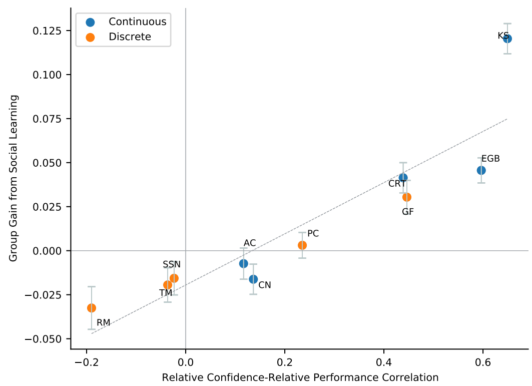
The reason is that peakedness captures only the pre-learning phase. In a social learning framework, what matters equally is whether participants can recognize a better answer once it is shown to them. CRT illustrates this clearly: although it has a very attractive incorrect answer, it also exhibits a large share of optimal switchers. Many participants fall for the intuitive wrong answer, but when presented with the correct one, they recognize it. A task with a negative β requires not only an attractive wrong answer but also a correct answer that is difficult to recognize as such once observed. This is related to the distinction between *insight* and *non-insight* problems in the psychology literature (Metcalfe and Wiebe, 1987; Kounios et al., 2006), though that classification also focuses primarily on the pre-learning stage and does not fully resolve the question.

Ex-Post Verifiability

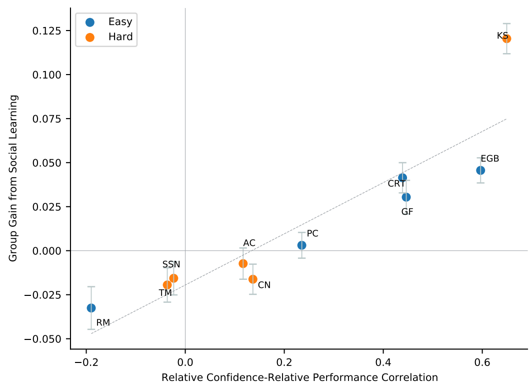
A more promising candidate is *ex-post verifiability*: the degree to which participants can directly compare two answers and assess which is more likely to be correct. The KS (knapsack) and CRT tasks are the clearest cases of ex-post-verifiable tasks in the set. In both, a participant shown the correct answer can, with some reflection, verify its superiority. Both tasks exhibit significant positive β and significant gains from social learning. By contrast, tasks such as RM and SSN involve probabilistic reasoning where the optimality of an answer is not directly verifiable from the answer itself, potentially explaining why participants shown the correct answer do not recognize it as such. While the evidence on this dimension covers only a subset of tasks and is therefore suggestive rather than conclusive, ex-post verifiability represents the most promising avenue for understanding cross-task heterogeneity in β.

Task Difficulty and Answer Type

Two further classifications (task difficulty and the type of available answers) do not appear to be predictive. Figure 13a splits tasks into discrete and continuous ones, and Figure 13b splits them into hard and easy based on the pre-learning optimality rate. Following the psychology literature on the hard-easy effect (Suantak et al., 1996; Moore and Cain, 2007; Moore and Healy, 2008), one might expect difficulty to affect confidence calibration and hence β. Neither split, however, explains the sign or magnitude of the relative confidence-relative performance correlation. Like peakedness, both classifications are properties of the pre-learning phase and do not capture what drives recognition of better answers during the learning phase.



(a) Scatter plot of relative confidence-relative performance correlation (on the x-axis) and the weighted net gain from social learning (on the y-axis). The error bars represent standard errors. Tasks are split into two groups, based on the type of required answer. "Discrete" tasks are the ones characterized by closed-ended questions, while continuous are the others. Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF= Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking.



(b) Scatter plot of relative confidence-relative performance correlation (on the x-axis) and the weighted net gain from social learning (on the y-axis). The error bars represent standard errors. Tasks are split into two groups of equal size, with the "Hard" tasks being the five tasks with the lowest pre-learning optimality rate and the "Easy" tasks being the five with the highest optimality rate. Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF= Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking.

7 Conclusion

Behavioral economics has documented an extremely rich set of behavioral biases, which have been shown to be impactful in laboratory and field settings. However, individuals do not make mistakes in isolation from others, but instead observe and learn from other individuals. It is unclear how observing others' actions and beliefs, that is social learning, impacts the incidence of behavioral biases. Does social learning reduce or amplify said biases?

Through a series of online experiments, I study social learning in a broad set of economically relevant tasks that reflect established behavioral biases. Crucially, this paper studies how social learning impacts group outcomes, that is the incidence of biased individuals among all participants. The evidence shows substantial heterogeneity in the effect social learning has on group outcomes, with the latter worsening for some biases and improving for others. This work also sheds light on the mechanism underlying this heterogeneity, showing how it is strongly predicted by the within-task relative confidence-relative performance correlation. The latter can be interpreted as a combination of the capacity of participants to assess the quality of their own actions (metacognition) and the quality of the answers from other participants, in a specific task. Together, these two findings imply that social learning is neither uniformly beneficial nor uniformly harmful: its impact depends on whether individuals who perform better are also better calibrated in their relative confidence assessments. This pattern is robust across substantially different experimental frameworks: it holds when social learning takes place in groups over multiple rounds, and is confirmed out of sample in an independent preregistered study in which participants are shown randomly drawn rather than contrasting answers.

Applications, Limitations and Further Avenues

The findings of this paper are relevant in any setting in which decision-makers are affected by behavioral biases and interact through social learning. A particularly relevant domain is non-institutional investment decisions, where investors routinely share assessments of new information with peers. Hong et al. (2004) show that social interaction among investors is associated with correlated trading behavior, consistent with the spread of common errors rather than their correction. More directly, Heimer (2016) finds that social interaction among retail investors is associated with worse trading performance, suggesting that peer learning may amplify rather than correct errors in this domain. The evidence presented in this paper helps explain when and why this occurs: social learning is most likely to spread errors when individuals who perform poorly are not well-calibrated in their relative confidence assessments, a condition that the experimental evidence suggests holds for biases closely related to the interpretation of noisy financial signals.
More broadly, these findings can help identify which biases are likely to persist or amplify in social environments, and which are instead naturally mitigated by peer interaction.

Two limitations of this work deserve mention. First, the relative confidence-relative performance correlation β reflects a composite object that may include a general individual-level tendency to trust or distrust others' answers, a domain-specific component reflecting differential ability in a given type of task, and an answer-specific component arising from the direct comparison of two concrete answers. Disentangling these components would require eliciting confidence both before and after the specific question and observed answer are revealed, and represents a natural direction for future research. Secondo, it is not clear whether all the abstract experimental tasks map cleanly into specific real-world applications, and the extent to which the mechanism generalises beyond the ten tasks studied here remains an open question. Third, the settings considered involve a relatively simple form of social learning, in which participants observe others' actions and beliefs without richer communication. Studying the impact of social learning on behavioral biases in settings with richer interaction structures, and documenting these mechanisms in more applied contexts, represent promising avenues for future research.

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Appendix A Additional Figures

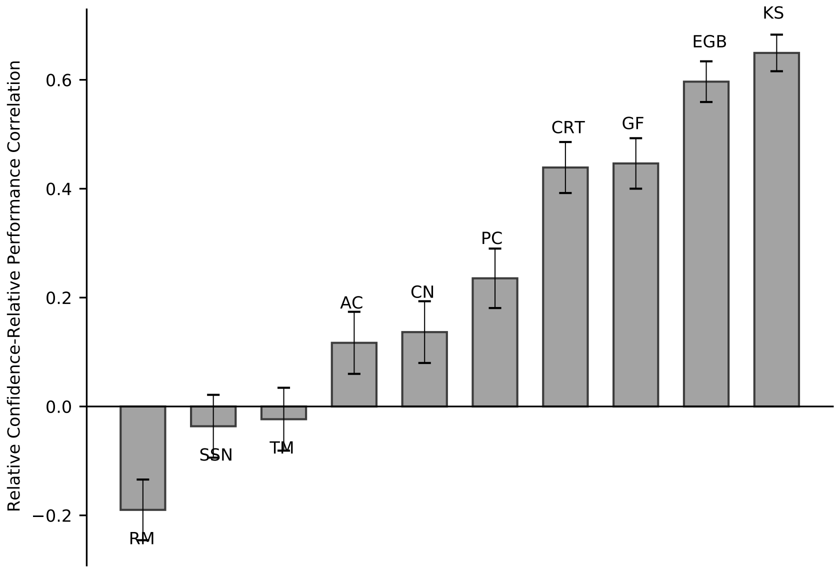


Figure A.1: Relative confidence-relative performance correlation (β), by task. Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF= Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking.

A.1 Unweighted Gains and Losses

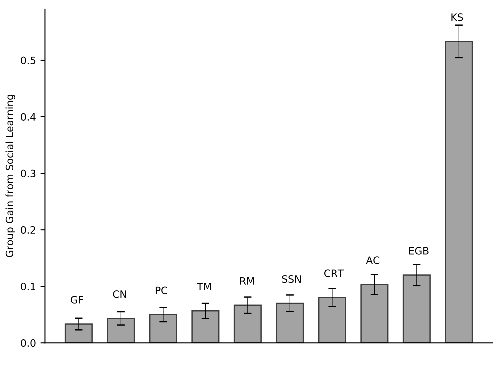


Figure A.2: Unweighted gains, by task. Error bars represent standard errors.

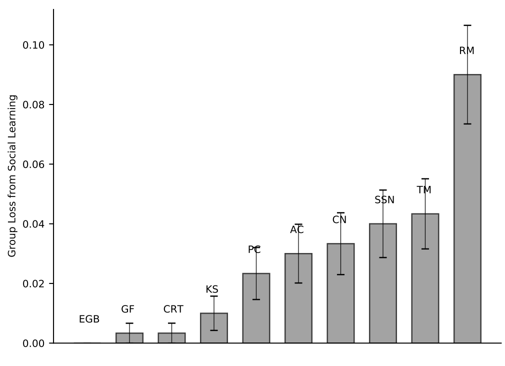


Figure A.3: Unweighted losses, by task (the bar is empty for EGB as there is no loss from social learning for that task). Error bars represent standard errors.

A.2 Answers Distribution

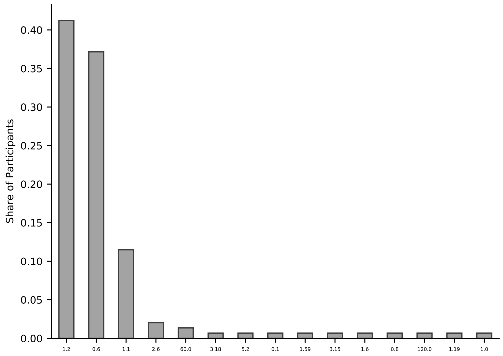


Figure A.4: Distribution of answers, CRT.

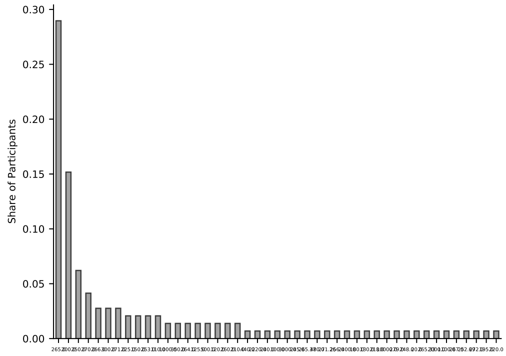


Figure A.5: Distribution of answers, EGB.

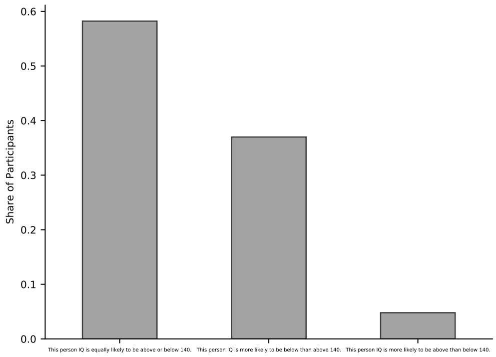


Figure A.6: Distribution of answers, RM.

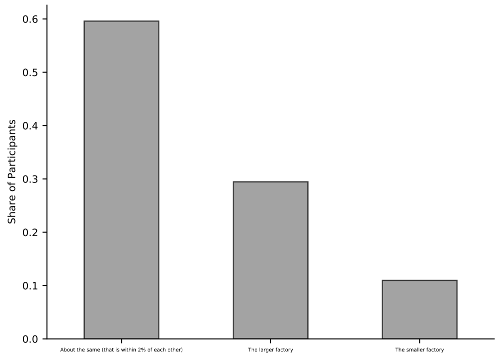


Figure A.7: Distribution of answers, SSN.

A.3 Gender Sample Split

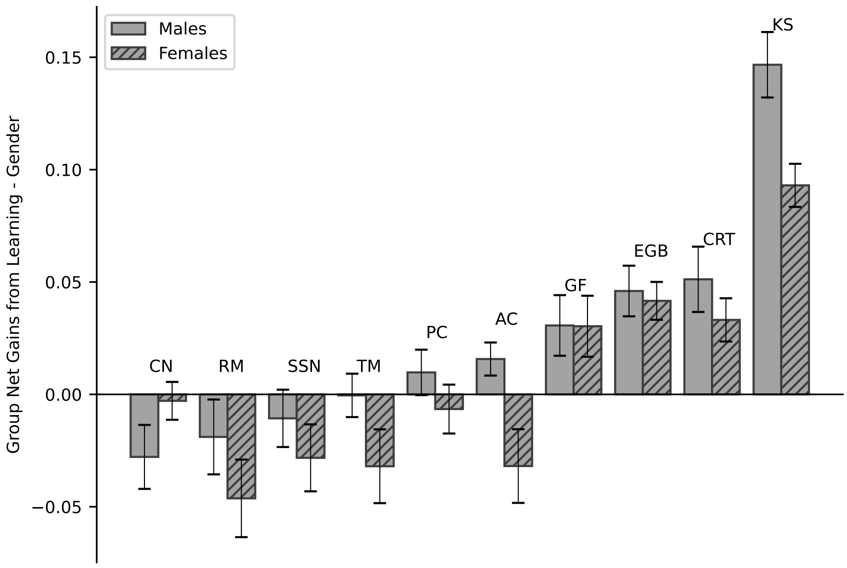


Figure A.8: Group gains from social learning, by task and gender. The weighted net gain measure, for each task, is built by weighting gains and losses with the optimality rate and its complement on 1, respectively. Error bars represent standard errors. Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF= Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking.

A.4 *OtherConf* Condition

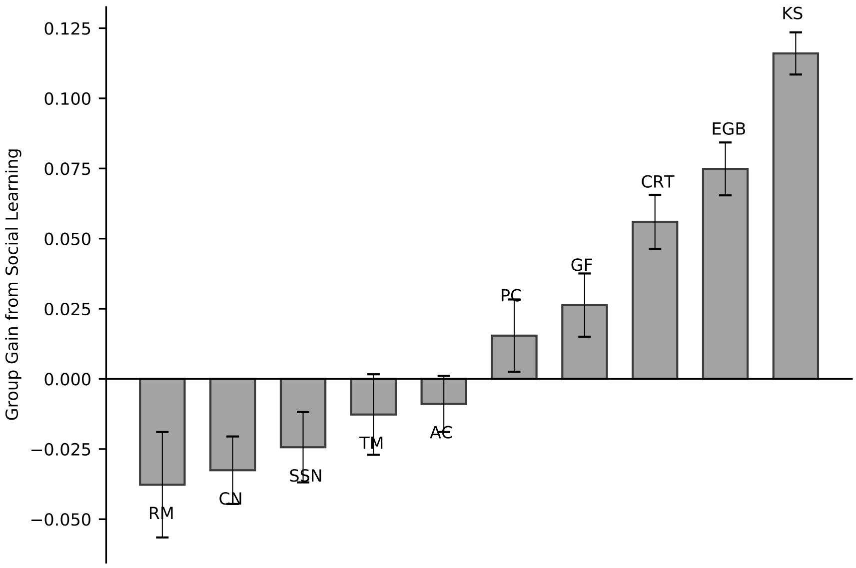


Figure A.9: Group gains from social learning, by task, for the OtherConf condition. The weighted net gain measure, for each task, is built by weighting gains and losses with the optimality rate and its complement on 1, respectively. Error bars represent standard errors. Task codes: AC=Acquiring-a-company; CN=Correlation neglect; CRT=Cognitive reflection test; EGB=Exponential growth calculation; GF= Predict sequence of draws; KS=Knapsack; PC=Portfolio choice; RM=Regression to the mean/Attribution; SSN=Account for sample size; TM=Marginal thinking.

A.5 Robustness

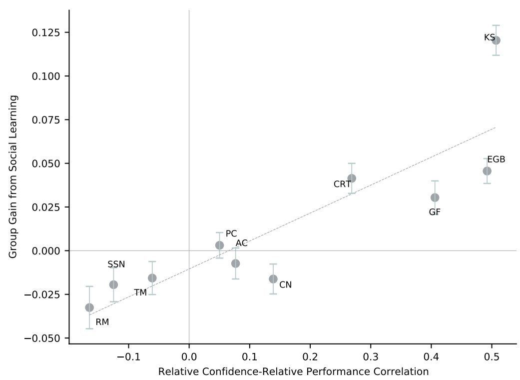


Figure A.10: Scatter plot of relative confidence-relative performance correlation (x-axis) and weighted net gain from social learning (y-axis). Relative confidence is encoded as a dummy ($c_{i,i} > c_{i,-i}$). Error bars represent standard errors. Task codes as in Figure A.1.

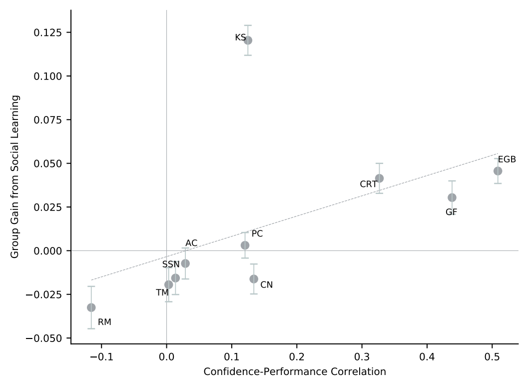


Figure A.11: Scatter plot of confidence-performance correlation (x-axis) and weighted group gain from social learning (y-axis). Error bars represent standard errors. Task codes as in Figure A.1.

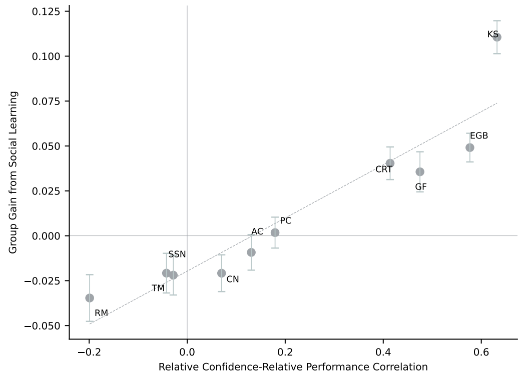


Figure A.12: Scatter plot of relative confidence-relative performance correlation (x-axis) and weighted group gain (y-axis), restricting to participants who did not express doubts about observed actions. Error bars represent standard errors. Task codes as in Figure A.1.

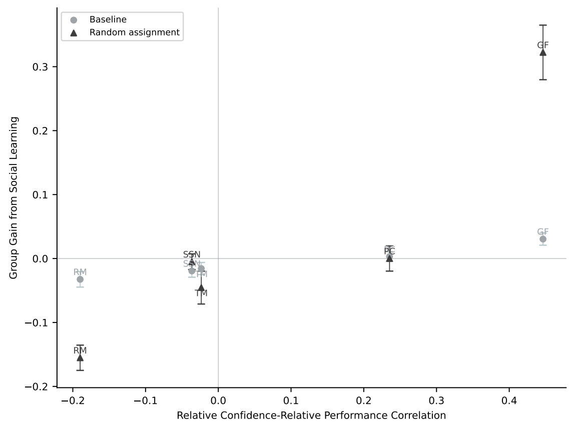
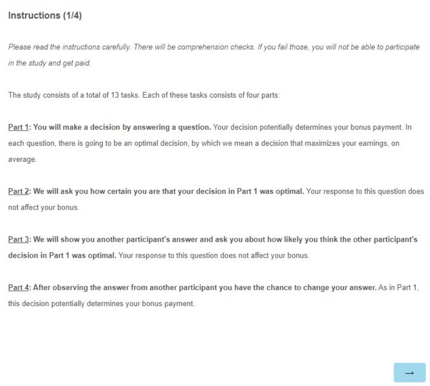


Figure A.13: Scatter plot of relative confidence-relative performance correlation (x-axis) and weighted group gain (y-axis). Circles: Baseline; triangles: robustness study with random answer assignment, restricted to observations where the shown answer differs from own answer. Error bars represent standard errors. Tasks: GF, PC, RM, SSN, TM.

Appendix B Proofs

Appendix C Experimental Instructions

C.1 General Instructions



Instructions (1/4)

Please read the instructions carefully. There will be comprehension checks. If you fail those, you will not be able to participate in the study and get paid.

The study consists of a total of 13 tasks. Each of these tasks consists of four parts:

Part 1: **You will make a decision by answering a question.** Your decision potentially determines your bonus payment. In each question, there is going to be an optimal decision, by which we mean a decision that maximizes your earnings, on average.

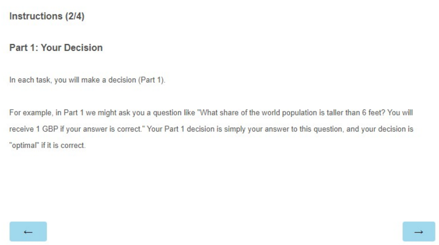
Part 2: **We will ask you how certain you are that your decision in Part 1 was optimal.** Your response to this question does not affect your bonus.

Part 3: **We will show you another participant's answer and ask you about how likely you think the other participant's decision in Part 1 was optimal.** Your response to this question does not affect your bonus.

Part 4: **After observing the answer from another participant you have the chance to change your answer.** As in Part 1, this decision potentially determines your bonus payment.

→

Figure C.1: Instructions screenshot 1



Instructions (2/4)

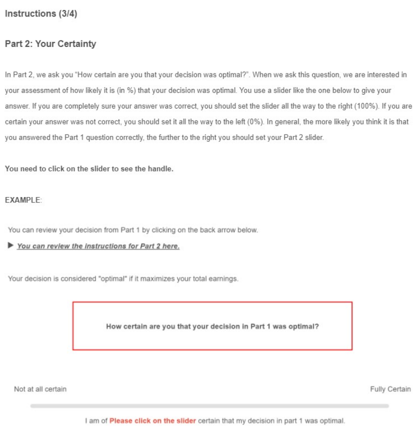
Part 1: Your Decision

In each task, you will make a decision (Part 1).

For example, in Part 1 we might ask you a question like "What share of the world population is taller than 6 feet? You will receive 1 GBP if your answer is correct." Your Part 1 decision is simply your answer to this question, and your decision is "optimal" if it is correct.

← →

Figure C.2: Instructions screenshot 2



Instructions (3/4)

Part 2: Your Certainty

In Part 2, we ask you "How certain are you that your decision was optimal?". When we ask this question, we are interested in your assessment of how likely it is (in %) that your decision was optimal. You use a slider like the one below to give your answer. If you are completely sure your answer was correct, you should set the slider all the way to the right (100%). If you are certain your answer was not correct, you should set it all the way to the left (0%). In general, the more likely you think it is that you answered the Part 1 question correctly, the further to the right you should set your Part 2 slider.

You need to click on the slider to see the handle.

EXAMPLE:

You can review your decision from Part 1 by clicking on the back arrow below.

▶ ***You can review the instructions for Part 2 here.***

Your decision is considered "optimal" if it maximizes your total earnings.

How certain are you that your decision in Part 1 was optimal?

Not at all certain | Fully Certain

I am of **Please click on the slider** certain that my decision in part 1 was optimal.

Figure C.3: Instructions screenshot 3

Part 3: Other Participants' Optimality

In Part 3, we show you another participant's answer to the **exact same question** and ask you "How likely do you think this participant's answer is optimal?". When we ask this question, we are interested in your guess of the probability that the shown answer is optimal. In general, the more likely you think it is that the shown answer is correct, the further to the right you should set your Part 3 slider.

You need to click on the slider to see the handle.

EXAMPLE:

A decision is considered "optimal" if it maximizes total earnings.

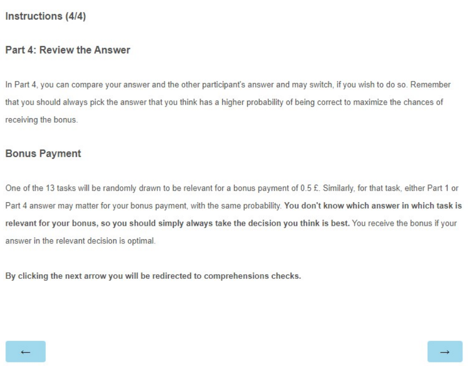
How likely do you think this participant's answer is optimal?

Extremely Unlikely | Extremely Likely

It is **Please click on the slider** likely that this participant's answers is optimal.

← →

Figure C.4: Instructions screenshot 4



Instructions (4/4)

Part 4: Review the Answer

In Part 4, you can compare your answer and the other participant's answer and may switch, if you wish to do so. Remember that you should always pick the answer that you think has a higher probability of being correct to maximize the chances of receiving the bonus.

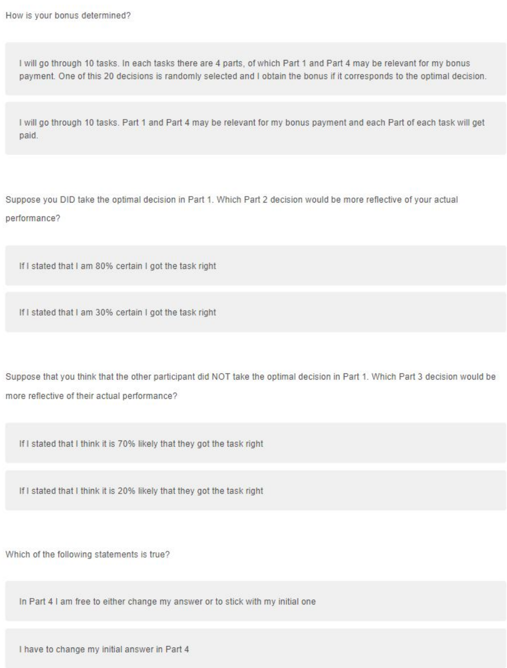
Bonus Payment

One of the 13 tasks will be randomly drawn to be relevant for a bonus payment of 0.5 £. Similarly, for that task, either Part 1 or Part 4 answer may matter for your bonus payment, with the same probability. **You don't know which answer in which task is relevant for your bonus, so you should simply always take the decision you think is best.** You receive the bonus if your answer in the relevant decision is optimal.

By clicking the next arrow you will be redirected to comprehensions checks.

← →

Figure C.5: Instructions screenshot 5



How is your bonus determined?

I will go through 10 tasks. In each tasks there are 4 parts, of which Part 1 and Part 4 may be relevant for my bonus payment. One of this 20 decisions is randomly selected and I obtain the bonus if it corresponds to the optimal decision.

I will go through 10 tasks. Part 1 and Part 4 may be relevant for my bonus payment and each Part of each task will get paid.

Suppose you DID take the optimal decision in Part 1. Which Part 2 decision would be more reflective of your actual performance?

If I stated that I am 80% certain I got the task right

If I stated that I am 30% certain I got the task right

Suppose that you think that the other participant did NOT take the optimal decision in Part 1. Which Part 3 decision would be more reflective of their actual performance?

If I stated that I think it is 70% likely that they got the task right

If I stated that I think it is 20% likely that they got the task right

Which of the following statements is true?

In Part 4 I am free to either change my answer or to stick with my initial one

I have to change my initial answer in Part 4

Figure C.6: Comprehension Questions.

C.2 Tasks

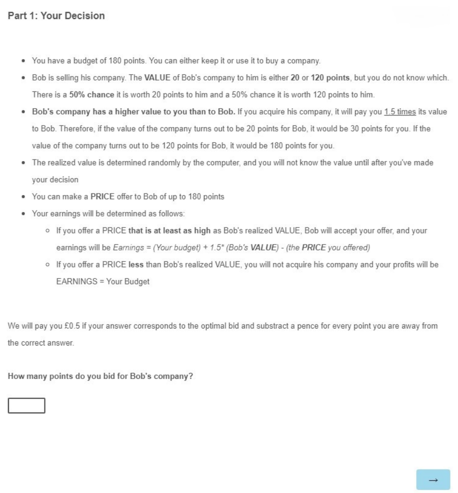


Figure C.7: Acquiring a Company (AC) instructions.

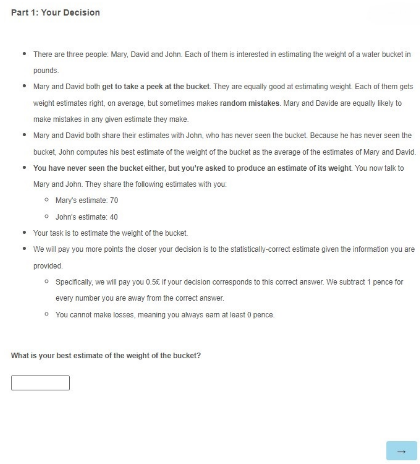


Figure C.8: Correlation Neglect (CN) instructions.

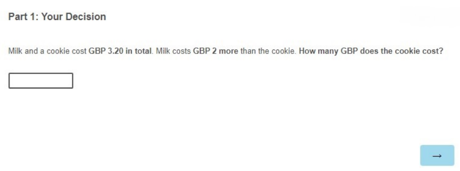


Figure C.9: Cognitive Reflection Test (CRT) instructions.

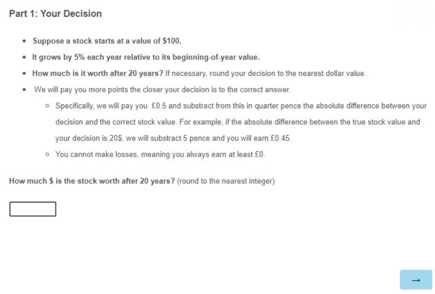


Figure C.10: Exponential Growth Bias (EGB) instructions.

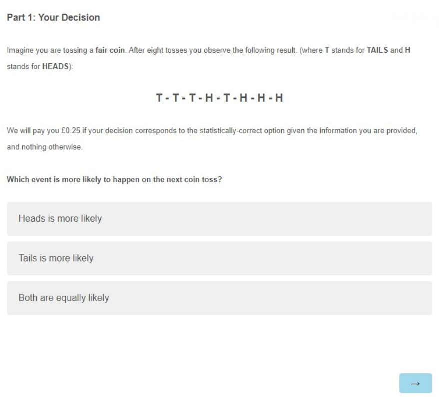


Figure C.11: Gambler's Fallacy (GF) instructions.

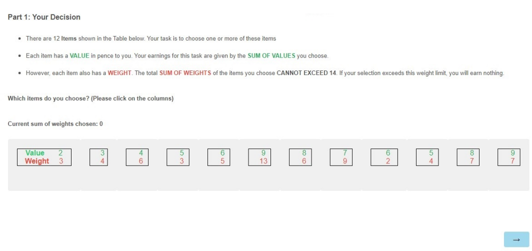
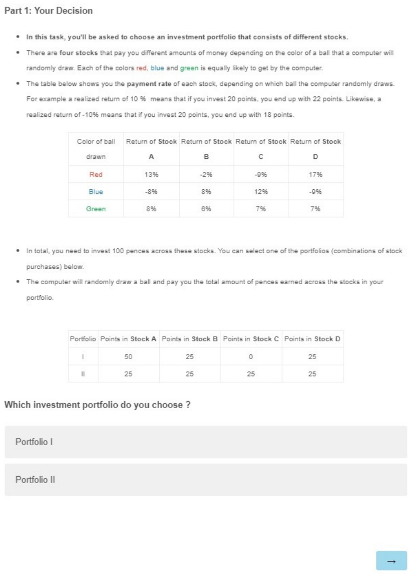


Figure C.12: Knapsack (KS) instructions.



Part 1: Your Decision

- **In this task, you'll be asked to choose an investment portfolio that consists of different stocks.**
- There are **four stocks** that pay you different amounts of money depending on the color of a ball that a computer will randomly draw. Each of the colors red, blue and green is equally likely to get by the computer.
- The table below shows you the **payment rate** of each stock, depending on which ball the computer randomly draws. For example a realized return of 10 % means that if you invest 20 points, you end up with 22 points. Likewise, a realized return of -10% means that if you invest 20 points, you end up with 18 points.

Color of ball drawn	Return of Stock A	Return of Stock B	Return of Stock C	Return of Stock D
Red	13%	-2%	-9%	17%
Blue	-8%	8%	12%	-9%
Green	8%	6%	7%	7%

- In total, you need to invest 100 pences across these stocks. You can select one of the portfolios (combinations of stock purchases) below.
- The computer will randomly draw a ball and pay you the total amount of pences earned across the stocks in your portfolio.

Portfolio	Points in Stock A	Points in Stock B	Points in Stock C	Points in Stock D
I	50	25	0	25
II	25	25	25	25

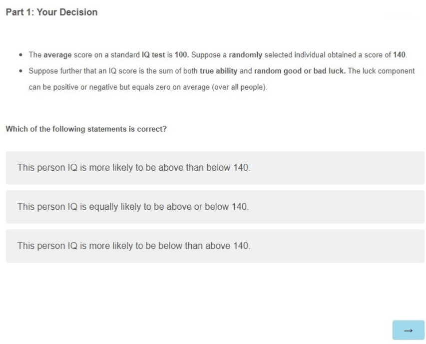
Which investment portfolio do you choose ?

Portfolio I

Portfolio II

→

Figure C.13: Portfolio Choice (PC) instructions.



Part 1: Your Decision

- The **average** score on a standard **IQ test** is **100.** Suppose a **randomly** selected individual obtained a score of **140**.
- Suppose further that an IQ score is the sum of both **true ability** and **random good or bad luck.** The luck component can be positive or negative but equals zero on average (over all people).

Which of the following statements is correct?

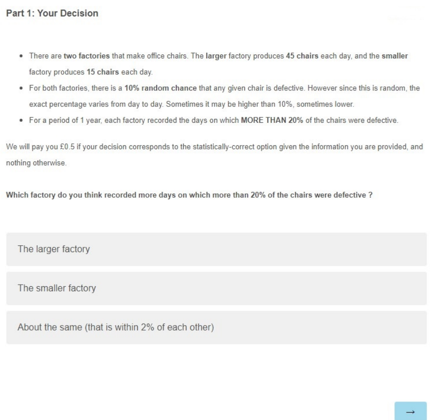
This person IQ is more likely to be above than below 140.

This person IQ is equally likely to be above or below 140.

This person IQ is more likely to be below than above 140.

→

Figure C.14: Regression to the Mean (RM) instructions.



Part 1: Your Decision

- There are **two factories** that make office chairs. The **larger** factory produces **45 chairs** each day, and the **smaller** factory produces **15 chairs** each day.
- For both factories, there is a **10% random chance** that any given chair is defective. However since this is random, the exact percentage varies from day to day. Sometimes it may be higher than 10%, sometimes lower.
- For a period of 1 year, each factory recorded the days on which **MORE THAN 20%** of the chairs were defective.

We will pay you £0.5 if your decision corresponds to the statistically-correct option given the information you are provided, and nothing otherwise.

Which factory do you think recorded more days on which more than 20% of the chairs were defective ?

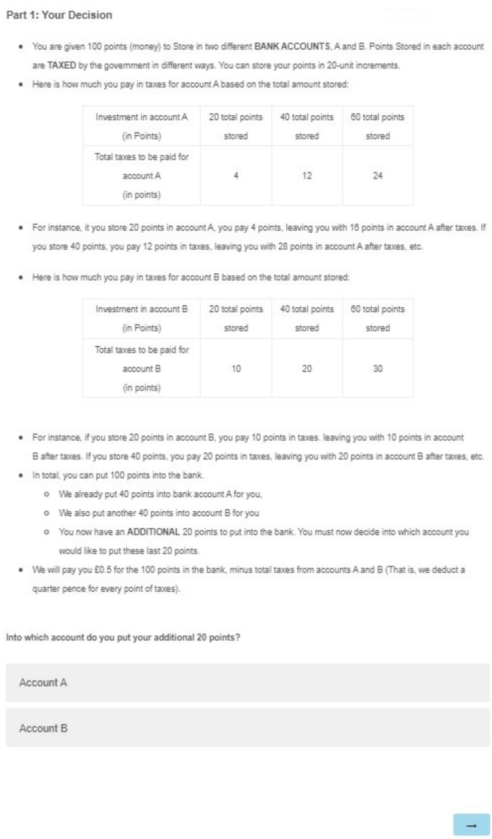
The larger factory

The smaller factory

About the same (that is within 2% of each other)

→

Figure C.15: Sample Size Neglect (SSN) instructions.



Part 1: Your Decision

- You are given 100 points (money) to Store in two different **BANK ACCOUNTS**, A and B. Points Stored in each account are **TAXED** by the government in different ways. You can store your points in 20-unit increments.
- Here is how much you pay in taxes for account A based on the total amount stored:

Investment in account A (in Points)	20 total points stored	40 total points stored	60 total points stored
Total taxes to be paid for account A (in points)	4	12	24

- For instance, it you store 20 points in account A, you pay 4 points, leaving you with 16 points in account A after taxes. If you store 40 points, you pay 12 points in taxes, leaving you with 28 points in account A after taxes, etc.
- Here is how much you pay in taxes for account B based on the total amount stored:

Investment in account B (in Points)	20 total points stored	40 total points stored	60 total points stored
Total taxes to be paid for account B (in points)	10	20	30

- For instance, if you store 20 points in account B, you pay 10 points in taxes. leaving you with 10 points in account B after taxes. If you store 40 points, you pay 20 points in taxes, leaving you with 20 points in account B after taxes, etc.
- In total, you can put 100 points into the bank.
  - We already put 40 points into bank account A for you,
  - We also put another 40 points into account B for you
  - You now have an **ADDITIONAL** 20 points to put into the bank. You must now decide into which account you would like to put these last 20 points.
- We will pay you £0.5 for the 100 points in the bank, minus total taxes from accounts A and B (That is, we deduct a quarter pence for every point of taxes).

Into which account do you put your additional 20 points?

Account A

Account B

→

Figure C.16: Thinking at the Margin (TM) instructions.

C.3 Tasks Description Table

Task	Short Description	Wrong Answer	Correct Answer
Correlation Neglect (CN)	Subjects estimate the weight of a bucket using guesses from Ann (70) and Charlie (40). They must infer Bob's guess was similar to Ann's, implying correlation.	55	40
Sample Size Neglect (SSN)	In a variant of the hospital problem, subjects judge which of two factories (small vs. large) more often exceeds a 20% defect threshold.	Equally likely	Smaller factory
Regression to Mean (RM)	Given a score of 140 on an imperfect IQ test, subjects judge whether true IQ is likely above or below 140, neglecting mean reversion.	Equally likely	More likely below 140
Acquiring a Company (AC)	A buyer offers to a seller whose value is 20 or 120. Value to buyer is 1.5x seller's. Seller accepts if offer $\geq$ value.	Offer $>$ 20	20
Cognitive Reflection Test (CRT)	"Milk and cookie cost £3.20. Milk costs £2 more. What's the cost of the cookie?" Measures override of intuitive response.	£1.20	£0.60
Knapsack (KS)	Choose highest-value subset of 12 items under a weight constraint. Tests optimization under constraints.	–	–
Portfolio Choice (PC)	Choose between portfolios of 4 assets. One follows naive 1/N heuristic, the other dominates.	1/N portfolio	–
Thinking at Margin (TM)	Allocate points between two accounts with different marginal and average tax rates. Choose based on marginal rates.	Lower average tax	Lower marginal tax
Exponential Growth Bias (EGB)	Stock grows at 5% annually for 20 years. Subjects often assume linear growth.	200 GBP	265 GBP

Table C.1: Task Description (adapted from Enke et al. 2022)